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CHAPTER 7

SYSTEM/COORDINATED TIMING

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CHAPTER 7. SYSTEM/COORDINATED TIMING

Progression is the movement of users along a designated route in a manner that minimizes stops.

Traditional coordination requires that cycle length, splits, and offsets be defined.

When intersections are close together, it may be advantageous to coordinate them to accommodate arriving platoons.

Instead of each signal operating independently, coordination allows signals to operate as a group, thereby synchronizing movements and allowing for better progression. Chapter 7 explains how the basic timing parameters introduced in Chapter 6 can be used in conjunction with coordinated features to operate a group of signals. In addition to providing guidance on the signal timing parameters (and typical values) that must be defined for coordination, Chapter 7 also highlights the advantages, disadvantages, and complexities associated with such a system.

Traditional coordination adds a layer of signal controller logic to the basic actuated logic described in Chapter 6. This system control method requires the practitioner to define a consistent cycle length for the corridor, as well as splits and offsets for each timing plan. Splits control the amount of time given to each phase in a cycle, and offsets control the time relationship between intersections. Both are described in detail in the following sections,

Traditional coordination will continue to be the dominant form of control for the foreseeable future, but some advanced systems (and likely future systems) will deviate somewhat from the traditional cycle/split/offset model. Some adaptive systems are able to maintain coordination without having to explicitly define cycle length, splits, and offsets; they are able to adjust timing values based on measured conditions. More information about advanced systems and features is available in Chapter 9.

7.1 APPLICATION OF A COORDINATED SYSTEM

The decision to use coordination should be influenced by a variety of factors, but most importantly, a practitioner should consider the operating environment, users, and appropriate user priorities. Coordination should be used to meet specific objectives; it is not appropriate in every situation. While coordination can reduce travel time, stops, delay, and queues for the coordinated movements, there may be consequences for the uncoordinated movements.

Information presented in the Federal Highway Administration (FHWA) report *Signal Timing on a Shoestring* (1) reveals that both simple and complex procedures can be used to identify which intersections to coordinate. In general, when intersections are close together (i.e., within one-half mile of each other), it is advantageous to coordinate them, particularly if volumes between the intersections are large. At greater distances (i.e., one-half mile or greater), the traffic volumes and potential for platoons should be further reviewed to determine if coordination would benefit system operations. If there is minimal traffic variation between intersections (e.g., caused by vehicles turning into or out of accesses), coordination at distances of a mile or more may be appropriate.

If corridor progression is the defined objective, the need for coordination can often be identified through observation of the traffic flow arriving from upstream intersections. If arriving traffic includes platoons that have been formed by the release of vehicles from an upstream intersection, coordination may provide desired progression benefits. On the other hand, if vehicle arrivals (a) tend to be random and are unrelated to the upstream intersection operations or (b) are broken up by accelerating/decelerating vehicles due to driveways or bus blockages, coordination may provide little benefit to the system.

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The costs and benefits of using coordination should also be weighed carefully against a community's multimodal goals. Many of the complaints from citizens related to the use of coordination are about the added delay and lack of responsiveness to their demand, particularly when there is no demand on conflicting movements. Delay for minor movements will be further exacerbated if the selected cycle length is unnecessarily long or the coordination plan is operating when traffic volumes are lower than is typical (e.g., during holidays that fall on a weekday).

7.2 COORDINATION PLANNING USING A TIME-SPACE DIAGRAM

As mentioned previously, traditional coordination uses the cycle/split/offset method to progress traffic. Various software and manual tools can be used to optimize this combination of signal timing parameters, but the practitioner should always ensure that the "optimized" results reflect the outcome based process (described in Chapter 3). (Typical software programs have a focus on **system** vehicle delay, which may **not** be an appropriate operational objective.) One tool that is often used to review coordination is the time-space diagram, which can be drawn manually or created using software. Basic elements of the time-space diagram are explained in this section, as well as how it can be used to evaluate coordinated timing plans.

Time-space diagrams are a visual tool that practitioners commonly use to analyze coordination strategies and modify timing plans. The diagrams illustrate the relationship among intersection spacing, signal timing, and vehicle movement. When used with some software programs, they can also be used to derive vehicle-based performance measures (also known as measures of effectiveness or MOEs), such as stops, vehicles arriving on green, and queue lengths. Coordinated parameter definitions and guidance are provided throughout the following sections, and many of them are explained using components of the time-space diagram in Exhibit 7-1.

7.2.1 Diagram Axes

Time-space diagrams are drawn with time on the horizontal axis and distance (from a reference point) on the vertical axis. In Exhibit 7-1, distance has been depicted using an aerial view of three signalized intersections. For ease of initially displaying relationships, the minor streets are all one-way (which is not typical). It is very important that time-space diagrams are drawn to scale so that an accurate relationship between time and distance is maintained for vehicle progression calculations.

The time-space diagram examples throughout this chapter assume that the corridor being analyzed is a north-south roadway. However, time-space diagrams can be used to analyze corridors traveling in any direction. The practitioner must simply assign a reference point for the distance axis. For example, an east-west corridor could be shown with the western-most intersection at the top of the vertical axis and the eastern-most intersection at the bottom.

7.2.2 Master and Local Clocks

Two types of clocks are running in the background during coordinated operations—the system master clock (i.e., field master, master controller, or central system) and the local (intersection controller) clock. The master clock is the background timing mechanism within the controller logic to which each local controller is referenced in

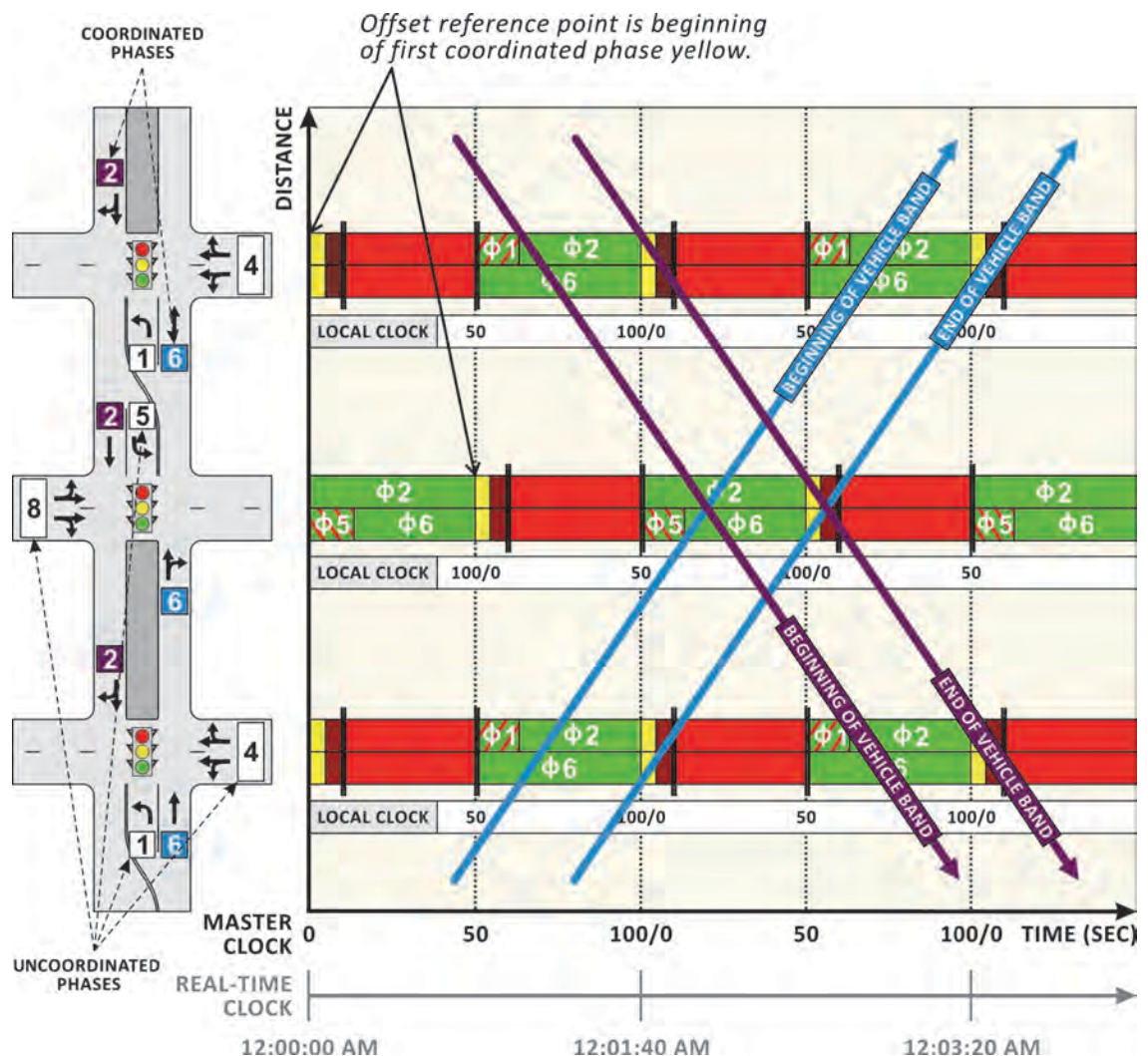
Time-space diagrams are a visual tool that can be used to assess coordination strategies and evaluate timing plans before field implementation.

The relationship between the master clock and the local (intersection controller) clocks is used to establish coordinated operations.

order to establish an offset between intersections (see Section 7.3.9). Both clocks must have the same time in order for coordination to work.

In the time-space diagrams in this chapter, the master clock is shown at the bottom of the diagram on the horizontal axis. The local clocks (if shown) are depicted below the ring-and-barrier diagrams associated with each intersection. In all exhibits in this manual, the offset reference point (see Section 7.4.8) is set to the beginning of first coordinated phase yellow (end of green), which is the beginning (zero point) of the local clock cycle. To help illustrate the relationship between the clocks, the master clock and local clocks in Exhibit 7-1 are shown in relation to a real-time clock below the horizontal axis. While the real-time clock shows times of day (note Exhibit 7-1 uses a sync reference of midnight, see Section 7.3.7), the master clock and local clocks show time relative to cycle length (in seconds).

Exhibit 7-1 Basic Time-Space Diagram



7.2.3 Signal Timing Graphics

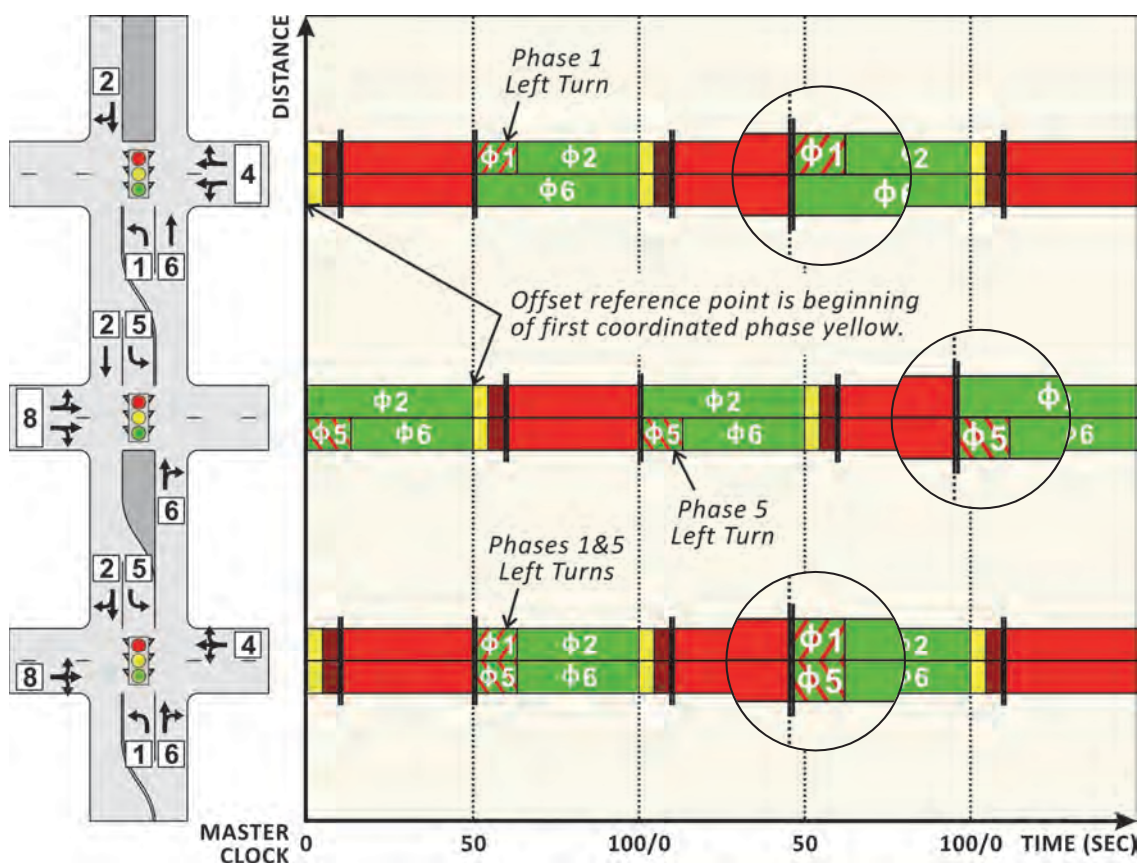
In the main portion of the diagram, signal timing information is provided for each intersection. Ring-and-barrier diagrams (explained in Chapter 5) depict the signal indications for the coordinated movements. In Exhibit 7-1, Phases 2 and 6 are the

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coordinated phases (which is typical but not an absolute requirement), and the ring-and-barrier diagrams show when those phases are receiving a green, yellow, or red indication. The dark red indicates the red clearance interval, and the bright red indicates when other uncoordinated phases (in this case, Phases 4 and 8) are receiving their green, yellow, and red clearance indications. Actual green times *may* differ from those shown in the ring-and-barrier diagrams because of gap outs and/or uncoordinated phases being skipped. Note that ring-and-barrier diagrams may show both rings (as in Exhibit 7-1) or a combined graphic that depicts both rings together (explained in detail in the following section).

7.2.4 Signal Timing Graphics for Left-Turn Phases

Signal timing information for left-turn phases (adjacent to the coordinated movements) is also depicted in a time-space diagram (see Exhibit 7-2). Protected left-turn movements are represented by hatching that is in the same direction as the through movement in the other ring. For example, with Phase 1 accommodating a northbound left-turn movement at the northern-most intersection, the diagonal hatching for Phase 1 is in the same direction as a northbound Phase 6 through vehicle trajectory. Because a southbound Phase 2 through vehicle trajectory would be in the opposite direction, a practitioner can easily identify that a southbound through vehicle would not be able to proceed through the intersection when Phase 1 is being served.



When two protected left-turn phases occur at the same time (as shown at the southern-most intersection of Exhibit 7-2), the hatching shows a period of time when

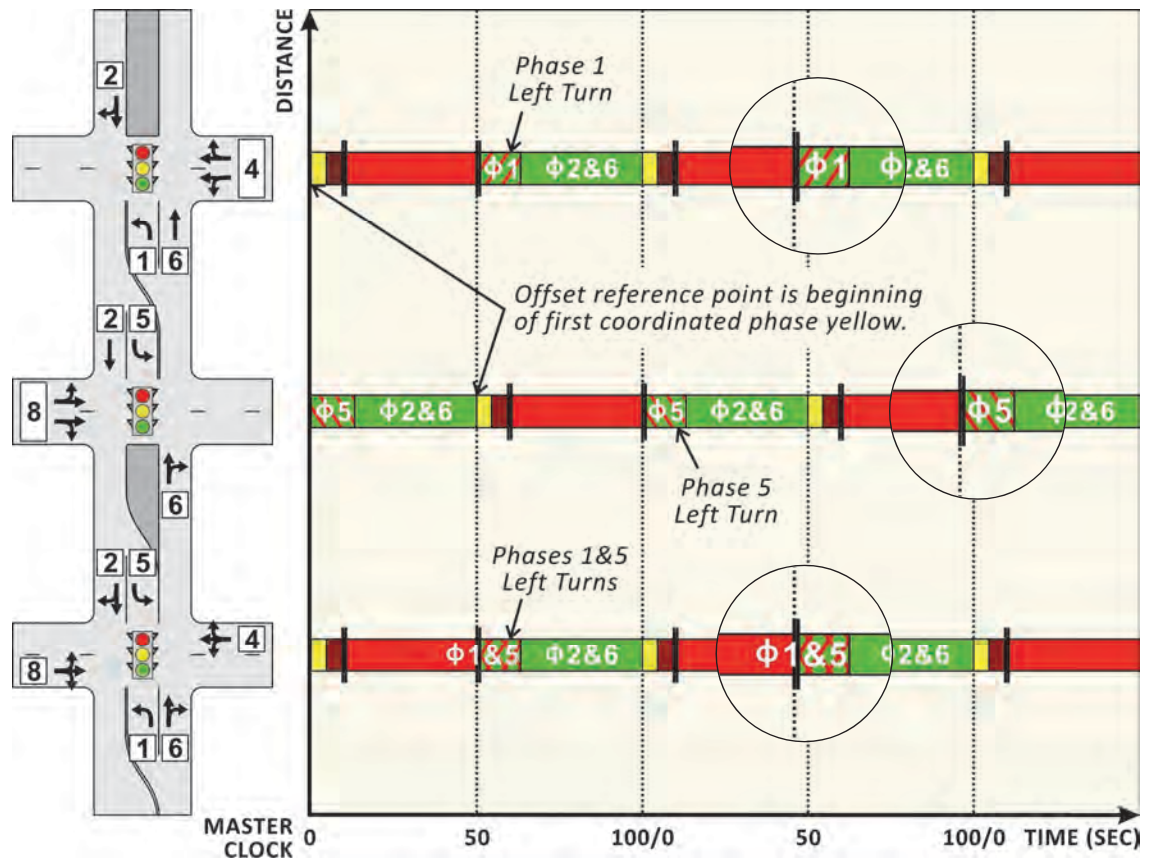
Ring-and-barrier diagrams graphically represent phases and their relationships at an intersection.

A vehicle trajectory is a representation of a vehicle traveling along the corridor that depicts both distance and time; it is explained in detail in the next section.

Exhibit 7-2 Time-Space Diagram Left-Turn Phasing (Separate Rings)

through movements are not possible in either direction. The two rings may not always be shown separately; in a combined ring-and-barrier diagram (see Exhibit 7-3), left-turn phases that occur at the same time are shown using a crisscross pattern (as highlighted at the southern-most intersection in Exhibit 7-3). These hatching conventions for left-turn phasing should make it easier for a practitioner to know at what point in the cycle a vehicle can progress.

Exhibit 7-3 Time-Space Diagram Left-Turn Phasing (Combined Rings)



7.2.5 Vehicle Trajectories

One of the most important components of a time-space diagram is the representation of vehicles traveling along the corridor. Vehicles are depicted using trajectory lines shown on top of the ring-and-barrier portion of the diagram. In Exhibit 7-1, they represent vehicles moving either north or south (toward the top or bottom of the diagram, respectively) as time increases (from left to right). (Note that if the distance axis represented an east/west street, the vehicle trajectories would depict eastbound/westbound vehicles.) Assuming that Exhibit 7-1 depicts a corridor with the northern-most intersection shown at the top of the diagram, the diagonal blue lines going from bottom/left to top/right represent northbound vehicles, and the diagonal purple lines going from top/left to bottom/right represent southbound vehicles. (Note that trajectory lines are not typically color-coded; blue and purple are used for example clarity only.)

The time-space diagram examples throughout this chapter show vehicle trajectories crossing the ring-and-barrier diagrams at the stop bar locations associated with the direction of travel (as shown in Exhibit 7-4). In other words, the northbound vehicles

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are drawn entering the intersections according to the signal indications at the bottom of the ring-and-barrier diagrams, and the southbound vehicles are drawn entering the intersections according to the signal indications at the top of the ring-and-barrier diagrams. With a combined ring structure (as shown in Exhibit 7-3), a thinner ring-and-barrier diagram can be used. Thinner ring-and-barrier diagrams make the “point of entry” conventions shown in Exhibit 7-4 of less significance. However, the remainder of this chapter uses ring-and-barrier diagrams with separate rings, in order to clearly demonstrate coordinated system concepts for both coordinated phases.

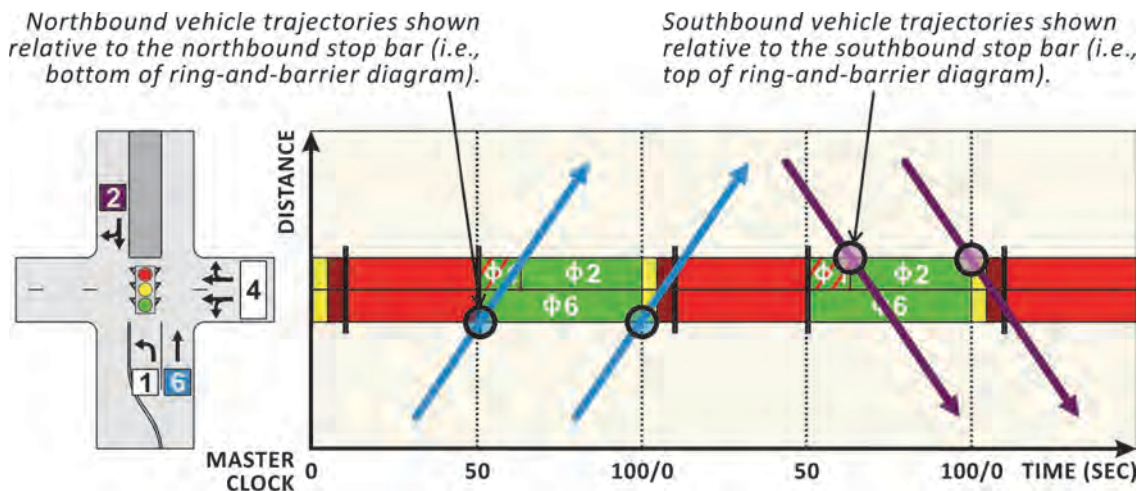


Exhibit 7-4 Vehicle Trajectory References

Because of the relationship between distance and time established in a time-space diagram, vehicle trajectories can be used to illustrate the speed that vehicles can progress along a corridor. The distance traveled by a vehicle divided by the elapsed time (distance divided by time, or change in y divided by change in x) represents the speed (see Exhibit 7-5). The assumed speed for coordination (i.e., the progression speed) may be the speed limit, the 85th-percentile speed, or a desired speed set by the practitioner.

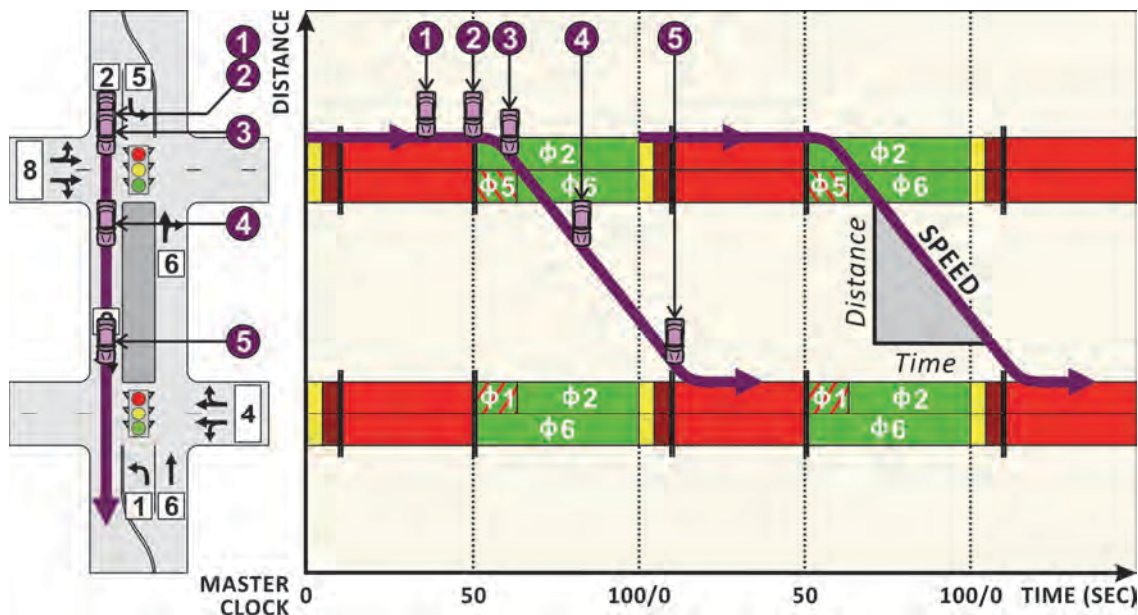


Exhibit 7-5 Vehicle Conditions on a Time-Space Diagram Trajectory

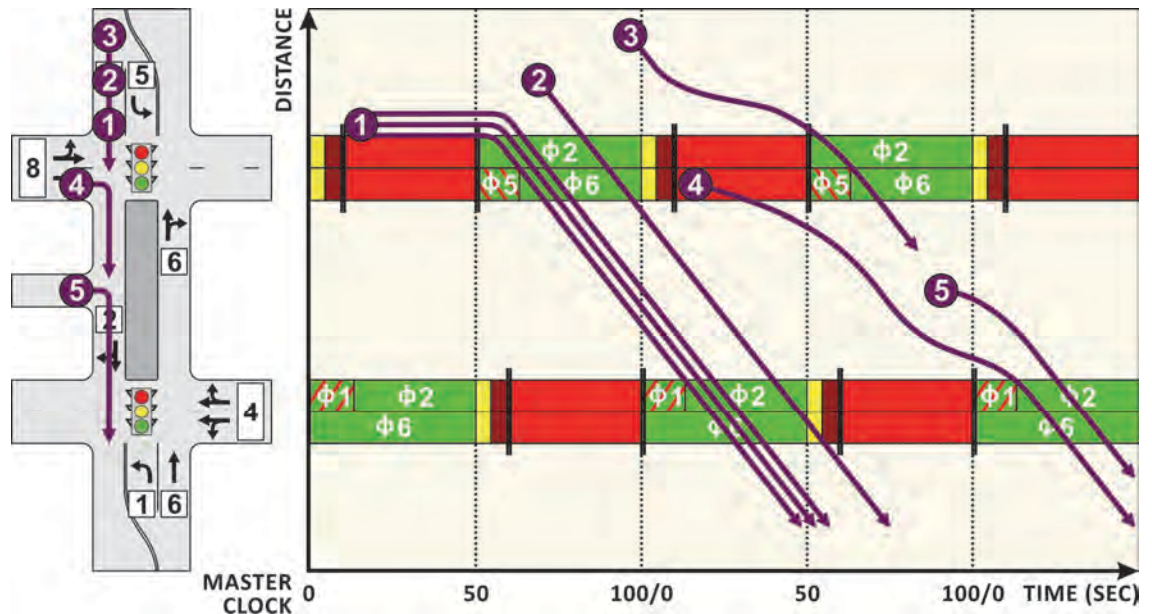
Exhibit 7-5 also illustrates the five different conditions a driver can experience when traveling between two signalized intersections. Each condition is described in detail, as it relates to the vehicle trajectory, in Exhibit 7-6. In this example, the vehicle trajectory shows that progression could be improved for the southbound movement by adjusting the offset (i.e., time relationship between intersections) so that the vehicle would not have to stop at both intersections.

Exhibit 7-6 Vehicle Conditions Associated with a Trajectory

No.	Condition	Description	Time-Space Diagram Representation
1	Stopped	Vehicle can be initially delayed as a result of a red indication at the signal.	Horizontal line; no change in distance as time moves forward.
2	Perception/Reaction Time	Drivers at the front of the queue must react to the green indication.	Horizontal line transitions to diagonal line; driver begins moving forward, increasing distance from the first intersection with time.
3	Acceleration	Vehicle accelerates up to the progression speed.	Diagonal line; vehicle moving forward with time.
4	Running Speed	Vehicle travels at the progression speed to the next intersection (often assumed to be the speed limit or an estimated progression speed).	Diagonal line; vehicle moving forward with time.
5	Stopping	Vehicle slows to stop due to red light (or stopped traffic).	Diagonal line transitions to a horizontal line.

Exhibit 7-7 shows a wider range of possible vehicle trajectories between two signalized intersections. Progression has improved over that in Exhibit 7-5, as vehicles can progress through both intersections (traveling at the progression speed) without stopping.

Exhibit 7-7 Example Vehicle Trajectories



The vehicle trajectories in Exhibit 7-7 (labeled 1 through 5) are described below:

1. Three vehicles travel from a stopped position at the first intersection through the downstream intersection.

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2. A vehicle traveling at the progression speed travels through both intersections without stopping.
3. A vehicle delayed at the upstream intersection (not coming to a complete stop) travels through the intersection upon the green indication.
4. A vehicle from the minor street travels to the downstream intersection. The vehicle enters the corridor when Phase 8 is receiving a green indication, which is a point in time when the coordinated phases are receiving a red indication. The vehicle has to slow as it approaches the downstream intersection, as shown by the more gradual slope of the trajectory.
5. A vehicle from a mid-block driveway travels through the downstream intersection.

7.2.6 Bandwidth

Section 7.2.5 stated that vehicle trajectories show the movement of vehicles as they progress along a coordinated corridor. However, there are select trajectories that illustrate vehicles traveling along a coordinated corridor without slowing or stopping. The first and last of these select vehicle trajectories outline the bandwidth (as shown in Exhibit 7-8). As the bandwidth gets wider, potential progression opportunities increase for vehicles traveling along the coordinated corridor. However, practitioners should always consider the system objectives. Wide bandwidths will only be effective if there are enough vehicles to utilize them.

Bandwidth represents progression opportunities and is illustrated on time-space diagrams as the shaded area between vehicle trajectories.

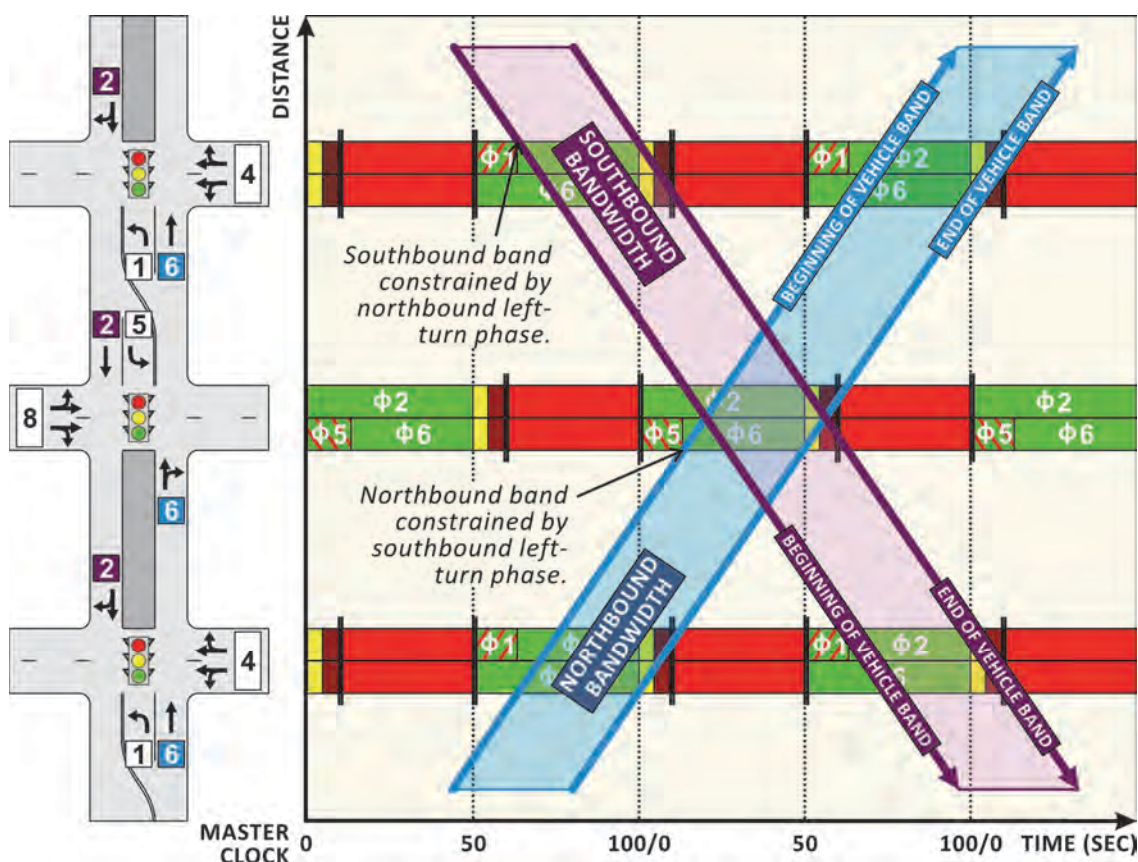


Exhibit 7-8 Bandwidth

Bandwidth is an *ideal* representation of progression, in that it does not explicitly account for vehicle acceleration from a stop, dispersion of vehicles as they travel from one intersection to the next, or queued vehicles at the downstream intersections. Actual bandwidth with actuated phases may vary due to phases gapping out (and not using their total split time). In addition, detector failure will limit the ability of bands to expand during light traffic. However, bandwidth is a useful tool for evaluating the relationship between coordinated parameters (including phase sequence, cycle length, splits, and offsets). A few important points to understand related to bandwidth are

- Bandwidth is dependent on the selected progression speed.
- Bandwidth can be different for each direction of travel.
- As additional intersections are added to a system, it is increasingly difficult to provide progression. It is sometimes better to break a long corridor into smaller segments (at convenient locations), particularly corridors with longer spacing between intersections. The segments can then be combined by selecting an offset relationship between the separately optimized segments, generally providing good progression in one direction. This approach is sometimes called a programmed stop. The stop location should be where there is adequate distance to provide storage without impacting the upstream intersection.
- During periods of oversaturated conditions (i.e., volume exceeds capacity), calculated bandwidth may never occur due to the influences of queuing. Strategies for mitigating oversaturated conditions are available in Chapter 12.
- Depending on the traffic volumes, timing plans that seek the greatest bandwidth may increase network delay and fuel consumption due to increased delay for uncoordinated movements.

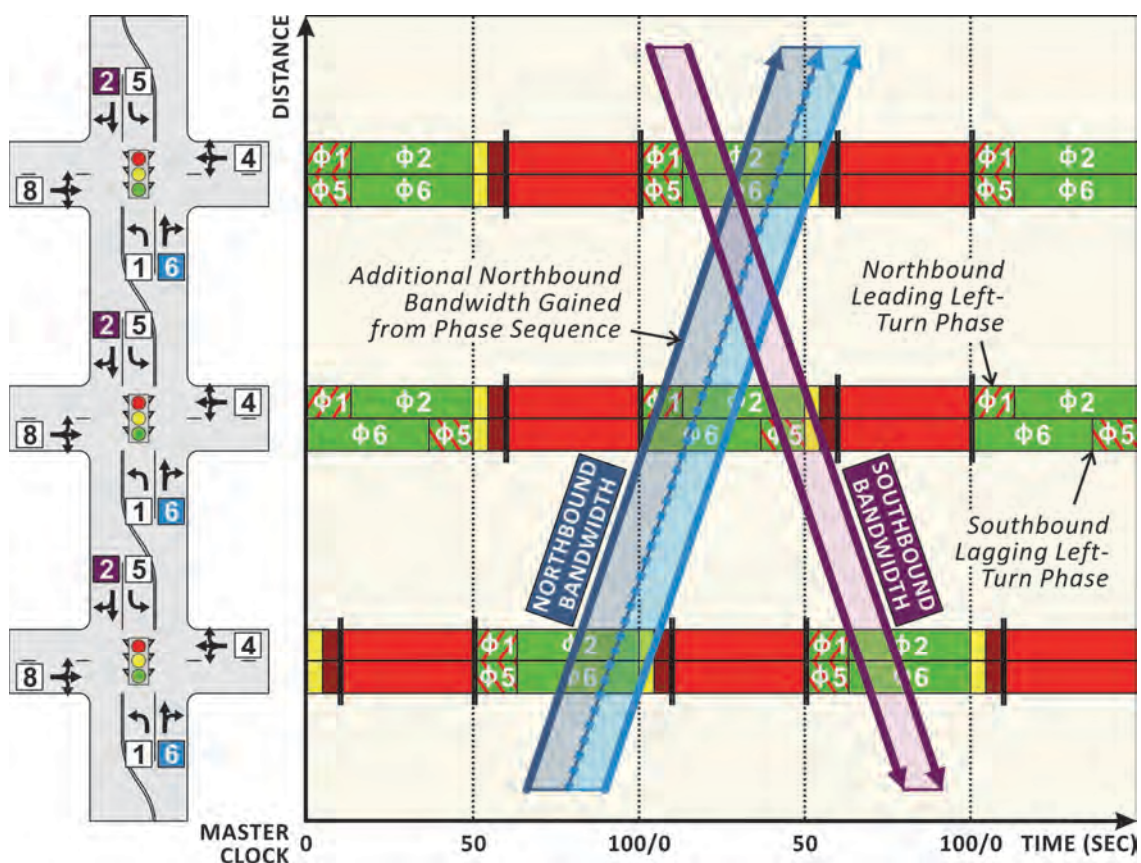
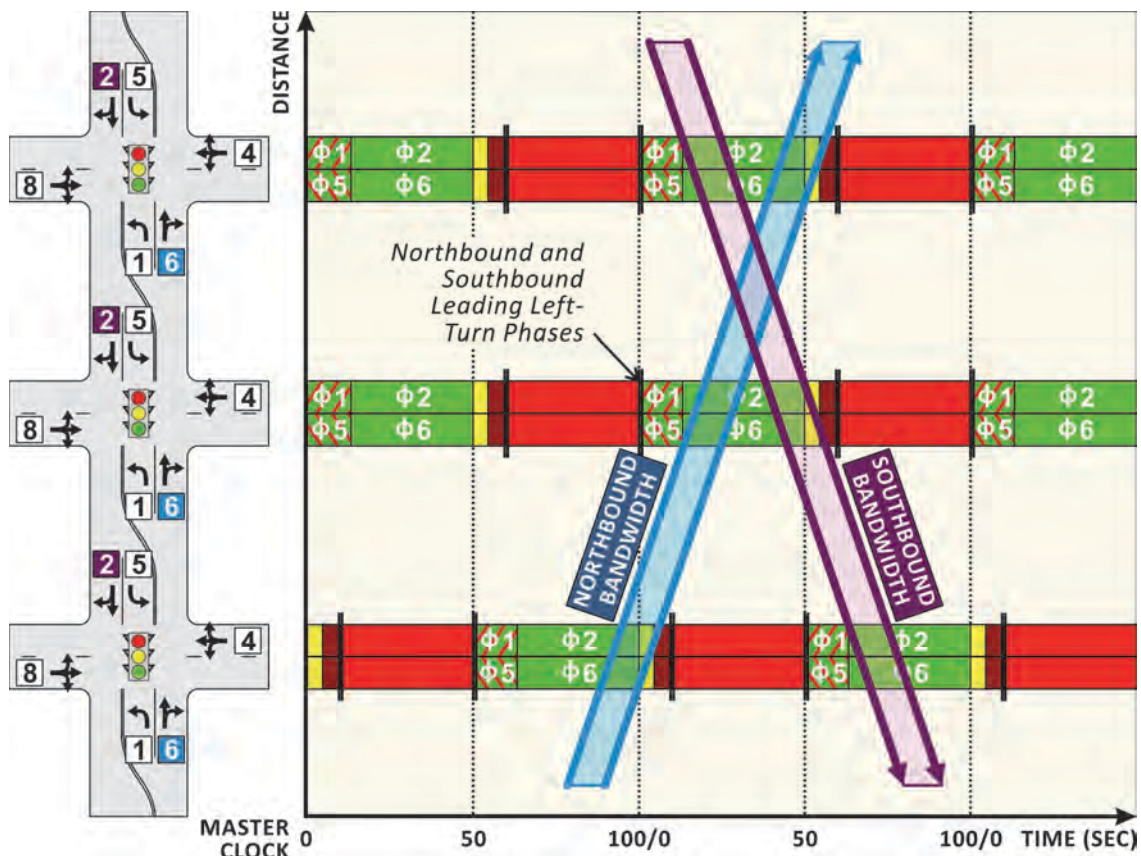
Phase sequence can have a significant impact on bandwidth and should be evaluated when intersections are not equally spaced or have different offsets. In particular, leading or lagging the left-turn movements can greatly influence the progression opportunities for the through movements. For example, Exhibit 7-9 shows a corridor with leading left turns, which results in fairly small northbound and southbound bands.

Exhibit 7-10 shows how the bandwidth can be increased for the northbound movement by using lead-lag phasing at the middle intersection. The northbound through movement is able to utilize the additional green time at the beginning of the cycle. While not shown in Exhibit 7-10, the southbound bandwidth could also have been increased by changing the phase sequence at the other two intersections. The practitioner is cautioned that lagging lefts are less efficient (see Chapter 5) and should only be used where the increase in coordination performance is significant.

The opportunity for progression may be further enhanced by using protected-permitted lead-lag phasing, which is feasible with flashing yellow arrows (FYAs). (For more information on FYAs and potential conflicts, refer to Chapter 4.) This technique can reduce the time necessary for the protected interval, which restricts movement on the opposing through phase. Protected-permitted lead-lag left-turn phasing is especially effective for coordinated signals where the progressed platoons in each direction do not pass through the intersection at exactly the same time (as is the case for the examples in Exhibit 7-9 and Exhibit 7-10).

Accommodating left-turning vehicles at signalized intersections is a balance among intersection safety, capacity, and delay.

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7.3 INTRODUCTION TO COORDINATION PARAMETERS

As mentioned previously, there are several fundamental parameters that must be selected or programmed in order for a traditional coordinated signal system to operate. This section describes these parameters, often using pieces of the time-space diagram introduced in Exhibit 7-1. Guidance on parameter **values** is provided in Section 7.4.

While this manual will generally use National Transportation Communications for ITS Protocol (NTCIP) “NTCIP 1202—Object Definitions for ASC” (2) terminology and data standards (i.e., format of inputs), it will also highlight when alternative terms or formats are used and may be preferred. It is important to understand that the NTCIP does not define how a controller works. It defines the **meaning** of parameters that can be used in any NTCIP compliant controller. For example, the NTCIP term “pattern” refers to a unique set of coordination parameters—cycle, split, offset, and sequence.

7.3.1 Coordinated Phases

Coordination requires the designation of a phase or multiple phases as the coordinated phase(s). Coordinated phases are selected so that traffic flow can be prioritized between multiple intersections, typically to reduce stops and delay for the coordinated movements. As mentioned previously, time-space diagrams focus on the coordinated phases and show signal indications and vehicle movements relative to those phases. While the examples in this manual focus on vehicle coordination, strategies to coordinate pedestrian, bicycle, and transit movements can also be implemented based on local desired outcomes.

Coordinated phases are distinguished from uncoordinated phases because they always receive a minimum amount of assigned green time every cycle. While it is possible to have an actuated portion of the coordinated phase(s) (a process that is discussed further in Section 7.5.2), there is always a non-actuated interval that is “guaranteed” every cycle. With a consistent cycle length, the guaranteed green interval can be used to maintain the coordinated relationship between intersections.

7.3.2 Cycle Length

Cycle length is the time required for a complete sequence of signal phases at an intersection. For an actuated control system, a complete cycle is dependent on the presence of calls on all phases. For a pretimed control system, a complete cycle can be recognized by a complete sequence of signal indications. In the time-space diagram shown in Exhibit 7-11, a cycle is depicted as one set of green, yellow change, red clearance, and red indications for the coordinated phases. The master clock (depicted on the horizontal axis) is shown relative to the programmed cycle length, which is 100 seconds in this example. As discussed in Section 7.2.2, the master clock is referenced to a real-time clock (typically at midnight or another time in the early morning).

All of the intersections included in a coordination plan should have the same cycle length, in order to maintain a consistent time-based relationship between intersections. One exception is an intersection that “double cycles,” serving phases twice as often as the other intersections in the system. In that case, the cycle length would be half the cycle length used at other intersections (as highlighted in Exhibit 7-11).

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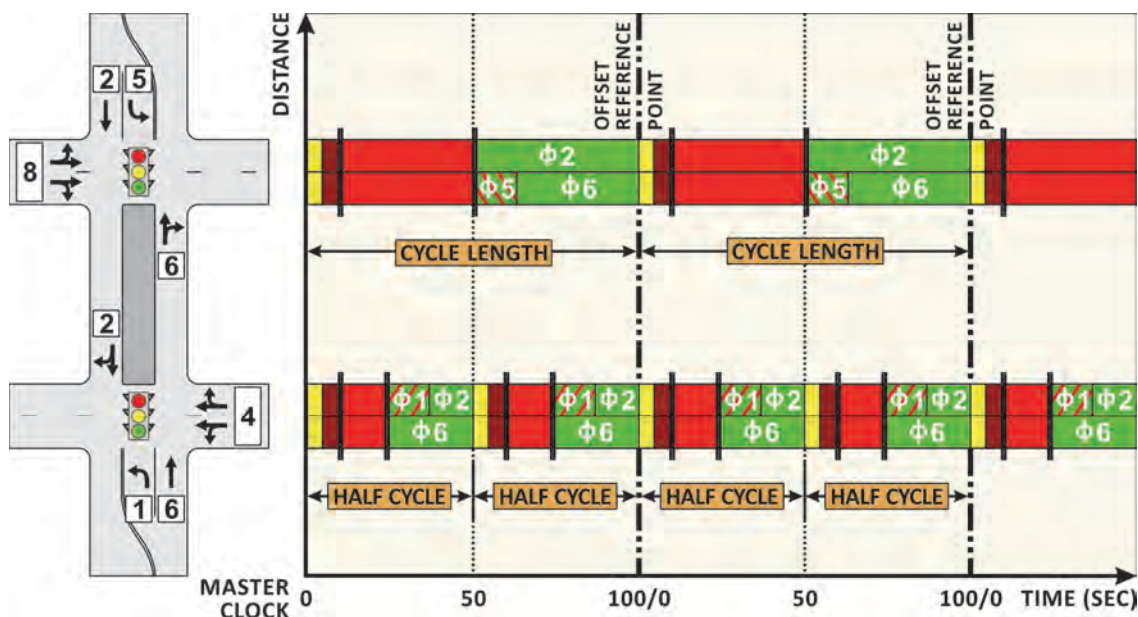


Exhibit 7-11 Cycle Length

7.3.3 Splits

Splits are the portion of the cycle allocated to each phase (green time plus clearances). Coordinated splits are selected based on the intersection phasing and expected demand. They can be expressed either in seconds (as outlined in NTCIP 1202, 2) or percentages of the cycle length. Split times include the green time, yellow change interval, and red clearance interval associated with a phase. Exhibit 7-12 highlights the splits associated with the coordinated phases. For implementation in a signal controller, the sum of the phase splits must generally be equal to (or less than) the cycle length if measured in seconds (or 100 percent if measured as a percent).

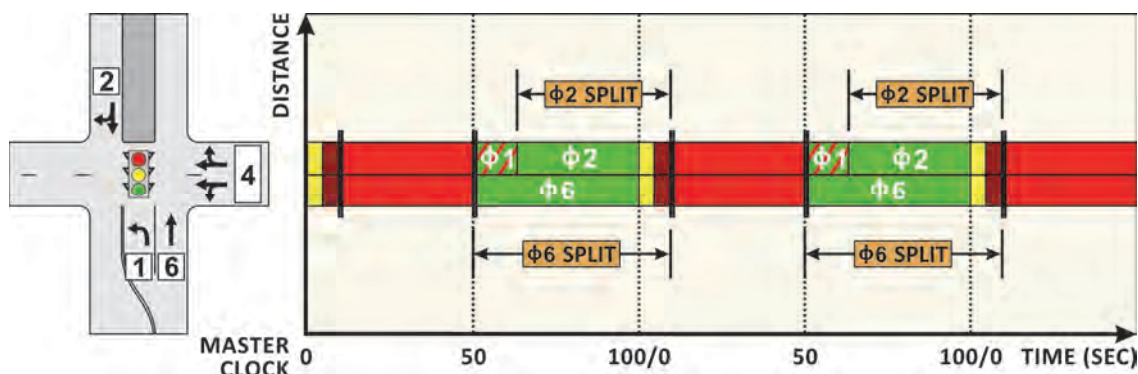


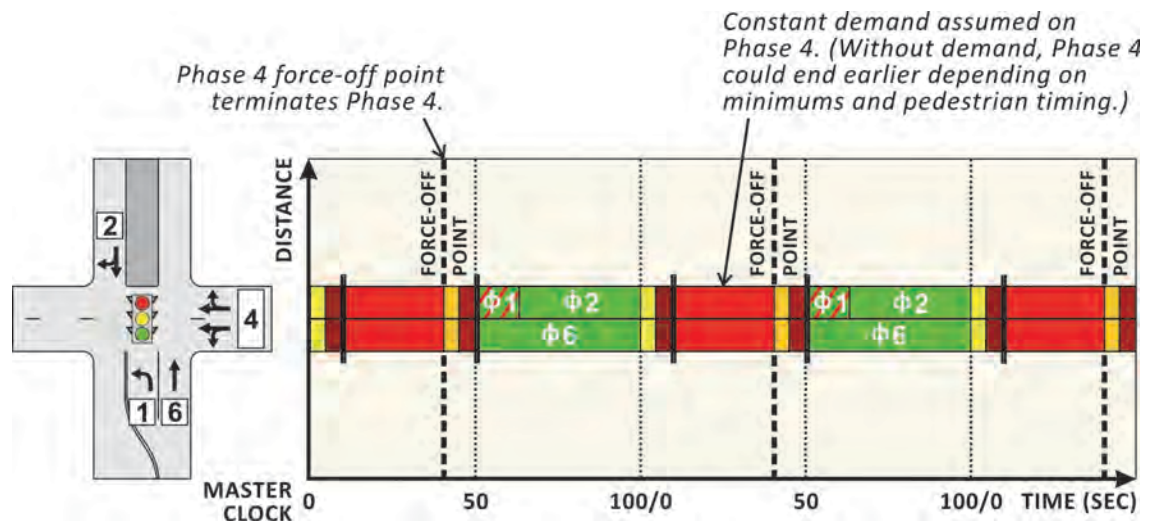
Exhibit 7-12 Splits

There are some controller firmware products that will allow splits to be shorter than pedestrian requirements. This is intended for situations in which there are few pedestrian calls. Full pedestrian times must always be served if called, so if splits do not include adequate time for pedestrian movements, the controller may need to transition in the next cycle to maintain coordination. More information on transition modes and pedestrian impacts is available in Section 7.5.

7.3.4 Force-Offs

Force-offs are used to enforce phase splits. They define the time during the cycle at which uncoordinated phases must end, even if there is continued demand, and ensure that control returns to the coordinated phase(s) no later than the programmed time, as illustrated in Exhibit 7-13. (Note that Phases 4 and 8 are shown in red because the time-space diagram highlights the signal status of the coordinated corridor.) Most modern controllers calculate force-off times based on the programmed split times (or split percentages). It should be noted that force-offs cannot override minimum times or clearance intervals. If force-offs are violated (i.e., the phase continues to serve minimum times or clearance intervals), the controller may have to re-sync to get back into coordination, unless following uncoordinated phases can give up time to allow the coordinated phase(s) to be served at their assigned time.

Exhibit 7-13 Force-Offs



There are two types of force-offs—floating and fixed—that determine how unused, actuated green time on the uncoordinated movements is shared with subsequent movements (as shown in Exhibit 7-14). **Floating** force-offs limit a phase to the assigned split time that was programmed into the controller. If a phase does not use all of its allocated time, then that movement cannot give the extra time to the next uncoordinated phase(s); all extra time is inherited by the coordinated phase(s). A **fixed** force-off maintains a phase's force-off point within the cycle. If a previous uncoordinated phase ends early, any following phase may use the extra time up to that phase's force-off point.

Exhibit 7-14 contrasts floating and fixed force-offs (in a single-ring example with Phase 2 being the coordinated phase). The top row illustrates a scenario in which demand equals or exceeds the allotted green time for all movements, and each phase is terminated at its respective programmed force-off point (sometimes referred to as “maxing out”). In other words, there is enough demand that none of the uncoordinated phases (Phases 3, 4, and 1) are able to end early. Note that instead of a force-off point, Phase 2 ends at a yield point. Yield points are explained in Section 7.3.6.

The next two rows illustrate the floating and fixed force-off concepts using a scenario in which there is unused green time on the uncoordinated movements. To better illustrate the difference between the two concepts, the demand affecting each

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phase (in seconds) is shown directly above each ring-and-barrier diagram. In this case, Phase 4 desires 30 seconds when its split time is 25 seconds. Under fixed force-offs, Phase 4 gets more time because Phase 3 only used 10 seconds of its allocated 25 seconds. Under floating force-offs, Phase 4 can never inherit any of Phase 3's unused time.

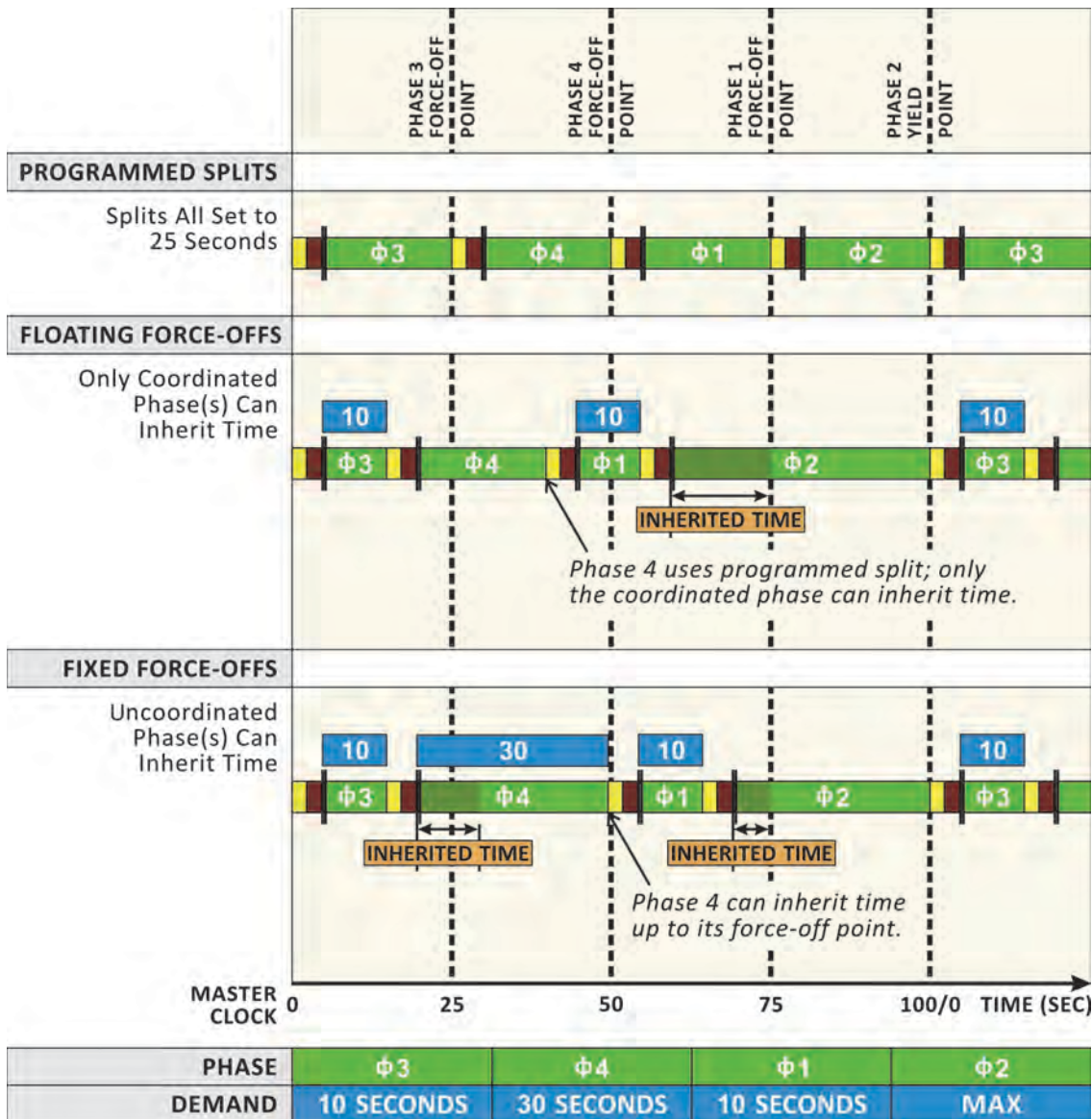


Exhibit 7-14 Fixed and Floating Force-Offs

7.3.5 Permissives

Permissives are windows of time when the controller can leave the coordinated phase(s) to serve the uncoordinated phases. In other words, they define the earliest and latest points at which an uncoordinated phase can **begin** in each ring. When a permissive period is active, the first uncoordinated phase in the sequence with demand will begin (i.e., an earlier phase in the sequence without demand will be skipped). Once the controller yields the coordinated phase to an uncoordinated phase, all subsequent phases with calls will be served; permissives no longer have an effect on the current

Permissives tend to be firmware specific.

cycle once the coordinated phase has yielded. Permissive periods are influenced by both the walk mode (discussed in detail in Section 7.5.1) and the type of permissive period that is programmed.

A practitioner will not be able to define permissive periods in every controller; most controllers automatically calculate and apply the largest permissive period possible (based on the vendor-specific approach to permissives). Regardless of whether permissives are defined by the practitioner or the controller, there are two general methods by which they are applied: (1) **simultaneous** permissive periods that start all permissives at the same time and (2) **sequential** permissive periods that start one at a time (i.e., the second permissive opens when the first closes, the third permissive opens when the second closes). The advantage of simultaneous permissives, under light traffic volumes, is the ability to skip Phase 4 and call Phase 1 early (normal sequence being Phases 1, 2, 3 and 4) if the Phase 1 call comes before Phase 4. Understanding the differences between these two methods will help a practitioner understand how a controller works under very low traffic volumes.

7.3.6 Yield Point

The yield point is the beginning of the first (or all) permissive periods and essentially determines the earliest point at which the coordinated phase(s) can begin termination and transition to uncoordinated phases. A yield point exists for each coordinated phase in each ring for most controllers, and while its primary purpose is to define a point in time when the controller can begin to terminate the coordinated phase(s), it is also the point at which transitions to new time-of-day plans typically take place. If lead-lag left-turn phasing is used for the movements adjacent to the coordinated phases, care should be taken to understand sync reference (as described below).

7.3.7 Pattern Sync Reference

Pattern sync reference is the start of the master clock (as outlined in NTCIP 1202, 2) that occurs daily. As defined previously, the master clock is the background timing mechanism within the controller logic to which each local controller is referenced for time-based coordination. Pattern sync reference may occur at midnight (as shown in Exhibit 7-1) or other times during the early morning (e.g., 1:00 a.m. or 3:00 a.m.) as selected by the practitioner or defined by the firmware. A reliable time reference must be established (e.g., GPS-based time reference) because local controller clocks must keep the same time for coordination to be most effective (i.e., the clocks must be synchronized).

7.3.8 Offset Reference Point

Offset reference points help structure the relationship among coordinated intersections by defining the point in time (relative to the master clock) when the cycle begins timing. Reference points can be defined using a variety of points in a cycle, as shown in Exhibit 7-15 (which assumes Phases 2 and 6 are the coordinated phases). Many modern controllers allow several options including

- Beginning of **first** coordinated phase green,
- Beginning of **both** coordinated phase green,

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- Beginning of **first** coordinated phase flashing don't walk (FDW), and
- Beginning of **first** coordinated phase yellow (end of first coordinated phase green).

The distinctions of “first” and “both” for the green, yellow, and FDW indications are important if the coordinated phases are using different values or the sequence has lead-lag left turns. While any offset reference can be computed from another offset reference (using cycle length and split times), the use of different offset references in the same system will produce poor results. This manual uses the beginning of first coordinated phase yellow (i.e., end of first coordinated phase green) as the offset reference because it is generally the most readily observable in the field.

Offset reference points should be used consistently.

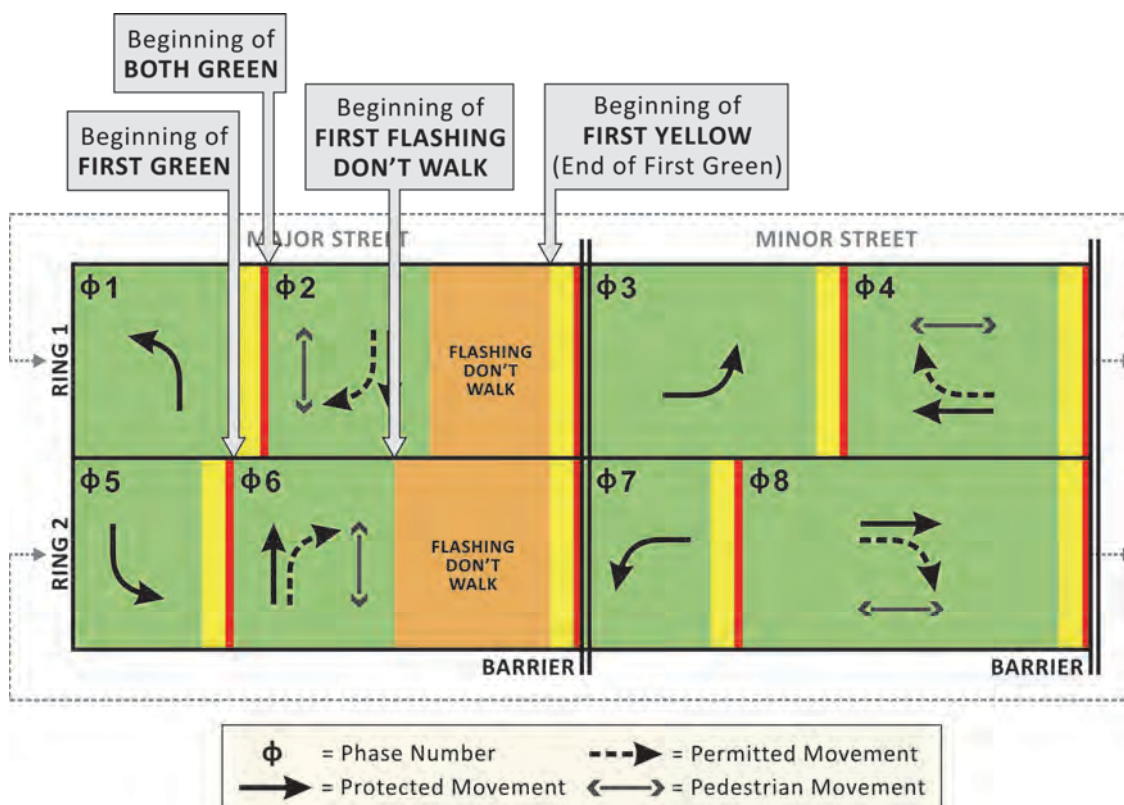


Exhibit 7-15 Offset Reference Points

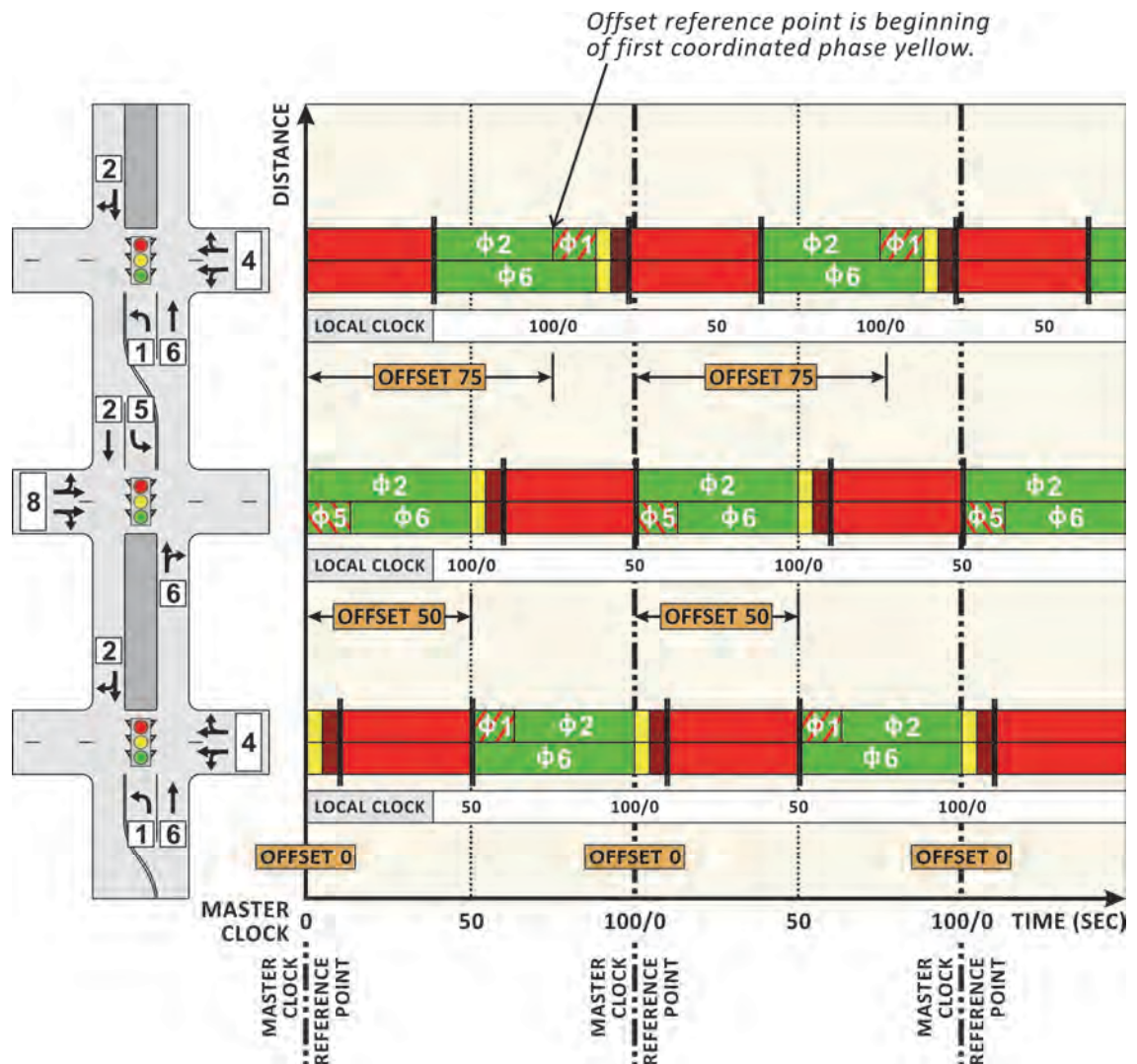
7.3.9 Offsets

Offsets define the time relationship (expressed in seconds or as a percentage of the cycle length) between the master clock and the coordinated phases at local intersections. Once an offset reference point has been identified, an offset is defined as the time that elapses between the beginning of the master clock cycle (master zero point) and the offset reference point at the local intersection (local zero point). Intersection offsets of 0 seconds, 50 seconds, and 75 seconds from the master clock are shown in Exhibit 7-16.

In Exhibit 7-16 (and throughout this manual), the offset reference point is the beginning of first coordinated phase yellow. Note the lagging left-turn sequence at the northern-most intersection. Because the offset reference point is the beginning of first coordinated phase yellow, the offset reference point occurs at the end of Phase 2. In this

example, the practitioner would see the Phase 2 yellow interval at the northern-most intersection begin 25 seconds after the Phase 2 (and Phase 6) yellow interval at the middle intersection.

Exhibit 7-16 Offsets



7.4 COORDINATION PARAMETER GUIDANCE

Once a practitioner has an understanding of the interactions and influence of coordinated timing parameters, he or she can use coordination planning tools (such as time-space diagrams) to select values for the coordinated parameters and create a timing plan. The following section provides additional guidance for this process.

7.4.1 Coordinated Phases Guidance

The desired outcome of coordination is often progressing the heaviest movements along a corridor, which frequently results in the major street through phases acting as the coordinated phases. (As explained in Chapter 5, these movements are typically designated as Phases 2 and 6.) However, the practitioner should explicitly consider traveler origin-destinations and traffic volumes when selecting the coordinated

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phase(s) because even a phase associated with a turning movement can be assigned as the coordinated phase.

7.4.2 Cycle Length Guidance

Cycle lengths, in combination with splits and offsets, establish relationships between intersections. Some cycle lengths will work better than others due to the time-space relationship between intersections, particularly if there is a desired system outcome (e.g., progression speed). Cycle lengths are frequently selected to address operations at a critical (or highest volume) intersection in a group of coordinated signals, but several additional cycle length selection methods are discussed in this section.

The practitioner should always consider whether intersections with long cycle length requirements, especially those with adequate separation from other intersections (i.e., no queue spillback to adjacent intersections), would better operate independently from a group. Cycle lengths that are too long may increase congestion, rather than reduce it, due to long wait times experienced by queued vehicles on both the minor and major streets. Longer cycle lengths tend to increase congestion when

- More vehicles move through an intersection than can be handled downstream.
- There are turn-lane storage bay issues. Long cycle lengths may cause vehicles in turn bays to back up into through lanes. In a similar manner, long cycles may cause through traffic to back up beyond turn-lane storage bays, restricting access for turning vehicles.
- Headways increase (reducing flow rate at the stop bar), as queued vehicles leave through lanes to enter turn lanes.
- There is increased variability in the actuated green times. Long cycle lengths can result in high variability in the green time used by the minor street, which may result in fewer vehicles arriving on green at the downstream intersection. This is particularly noteworthy when split times exceed 50 seconds (3).

7.4.2.1 Manual Methods for Cycle Length Selection

Although most signal timing plans are developed using computer-based analysis, the tools are only as good as the traffic data and the optimization model. Often, too much emphasis is placed on the tool and too little on understanding the basic relationships. In many simple networks with modest traffic volumes, timings can be developed manually by selecting an appropriate cycle length (and offsets) based on block spacing.

A manual method for determining cycle lengths in a traffic signal network was first developed in Los Angeles (the City). The City used a traffic signal timing strategy that was elegantly simple for its then master-planned robust grid system. Most of the traffic signals in the San Fernando Valley part of the City operated using just two phases and permitted left turns, with signals spaced every quarter-mile. A 60-second cycle length was selected, and the green time was equally distributed between the minor street and major street traffic. Given the timing parameters and intersection spacing, a vehicle traveling at 30 miles per hour (mph) would reach the next quarter-mile signal in 30 seconds, or half of a cycle length. This resulted in an “alternating” system of offsets

Cycle length selection is typically based on traffic data that are collected during representative time periods. If available, automated data collection provides more robust information for selecting cycle lengths.

between adjacent intersections (as shown in the time-space diagram in Exhibit 7-17). With 30-second splits, progression was achieved on both the major and minor streets.

Exhibit 7-17
Alternating System of Offsets

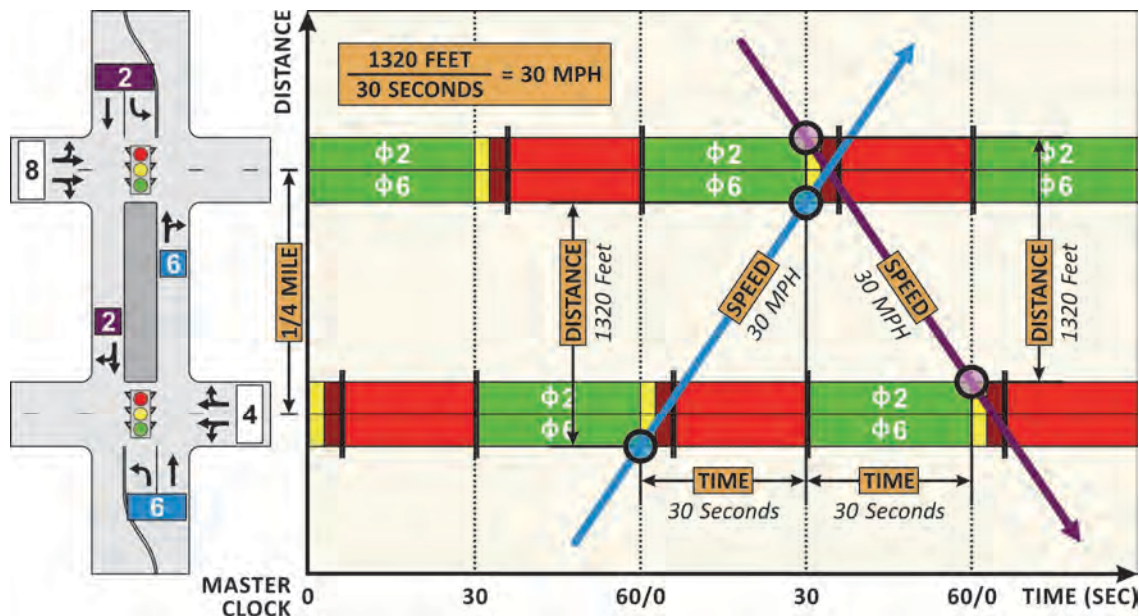


Exhibit 7-18 provides additional guidance on the relationship between cycle length, block spacing, and speed. The speed of progression can essentially be determined by dividing the block spacing by one-half, one-fourth, and one-sixth of the cycle length, respectively, for single, double, or triple alternate systems (depicted in Exhibit 7-19, Exhibit 7-20, and Exhibit 7-21).

Exhibit 7-18
Progression Speeds for Predetermined Cycle Lengths and Equal Block Spacing

Block Spacing (Feet)	Progression Speed (MPH)								
	60-Second Cycle Length			90-Second Cycle Length			120-Second Cycle Length		
	Single	Double	Triple	Single	Double	Triple	Single	Double	Triple
330	8	15	23	5	10	15	4	8	11
660	15	30	45	10	20	30	8	15	23
1320	30	60	90	20	40	60	15	30	45
2640	60	120	180	40	80	120	30	60	90

Note: Patterns that are grayed out are not recommended because of undesirable speeds.

In single alternate systems, every other intersection has an offset equal to half the cycle length relative to the master clock. In double alternate systems, every pair of intersections has a half-cycle offset relationship. In triple alternate systems, every three adjacent intersections have a half-cycle offset with the master clock. While triple alternate systems have relatively poor progression bands, the simultaneous offsets may be effective with high turning volumes from the minor streets. Minor street traffic would get an early green (explained in Section 7.6.2) at the middle intersection, potentially clearing queues before through traffic arrived.

Depending on the block spacing, desired progression speed, and type of alternate system, the practitioner should be able to determine an appropriate cycle length using the information in Exhibit 7-18. For example, if a practitioner wants to time signals along a corridor with quarter-mile spacing, using a single alternate system, for a desired

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progression speed of 30 mph, the signals should be timed using a 60-second cycle length.

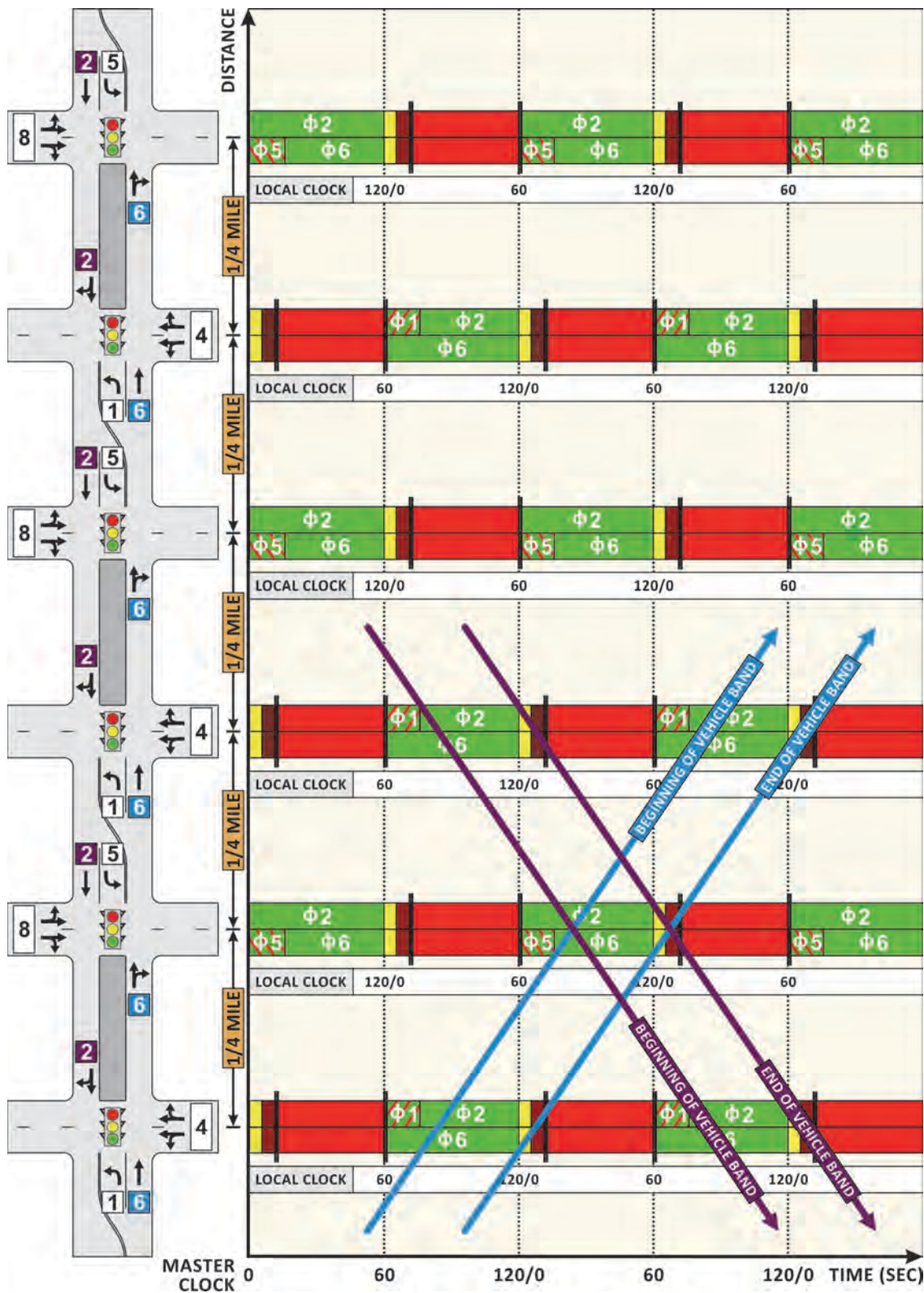
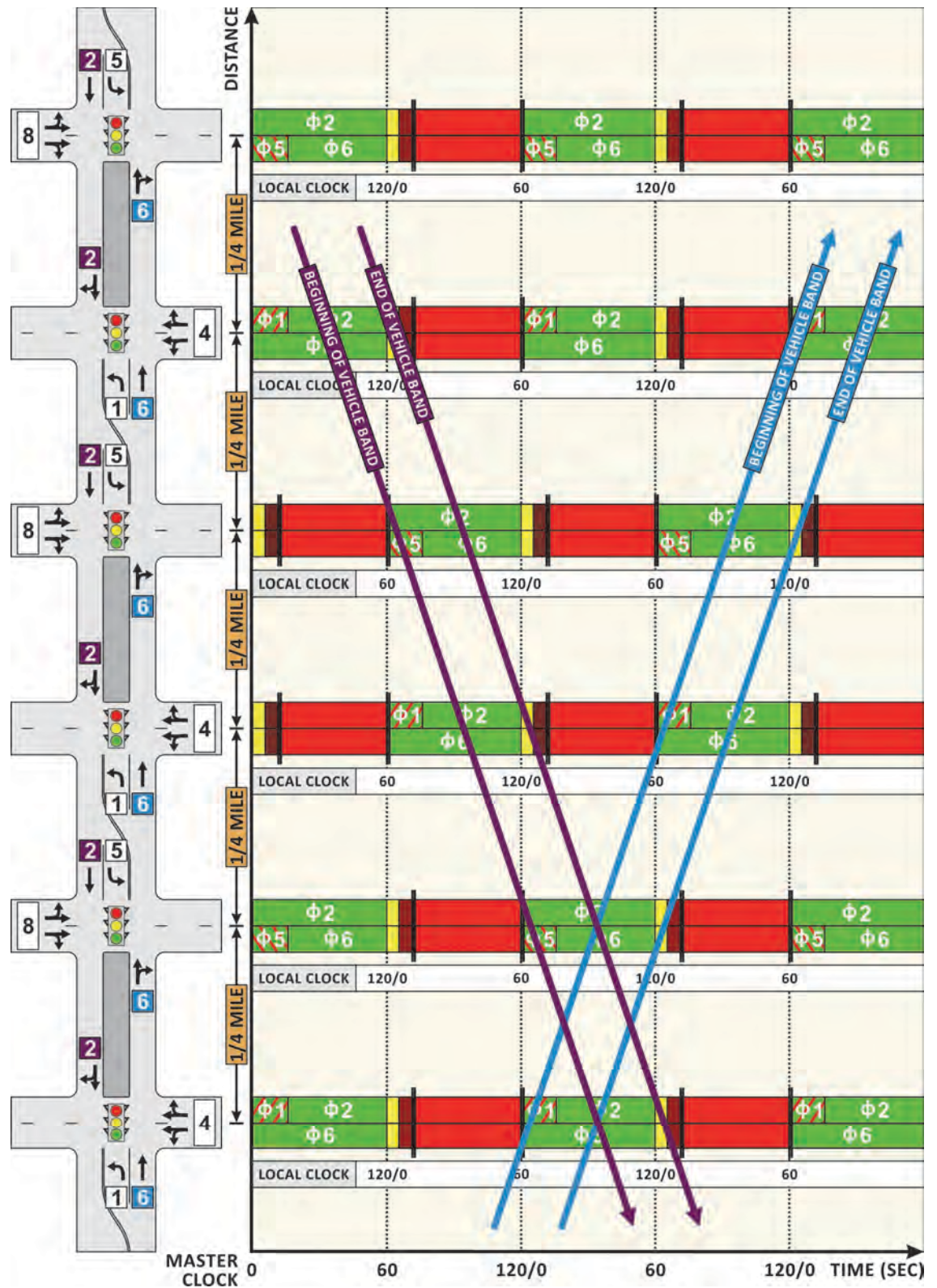


Exhibit 7-19 Single Alternate Coordinated System

Exhibit 7-20 Double Alternate Coordinated System



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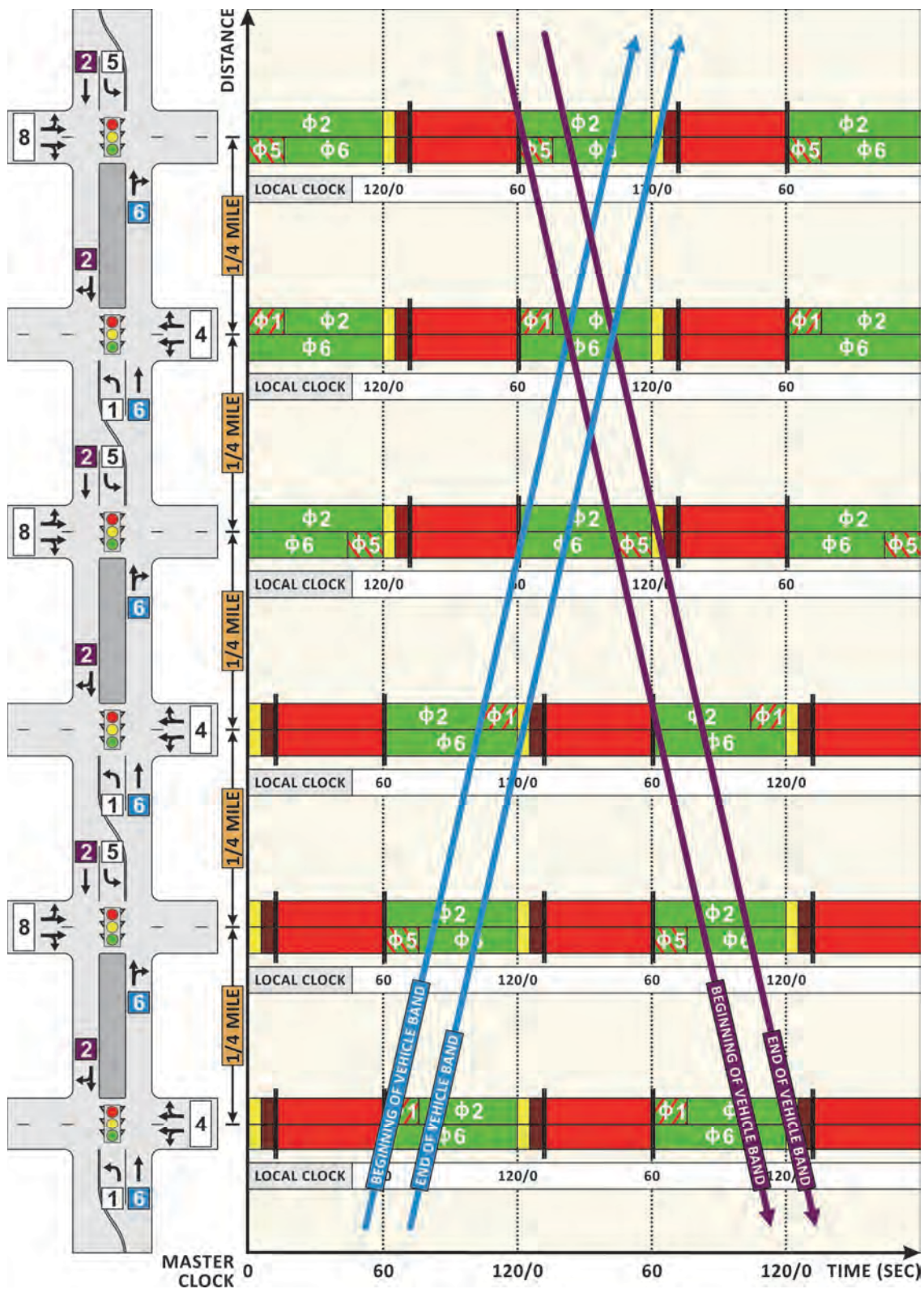
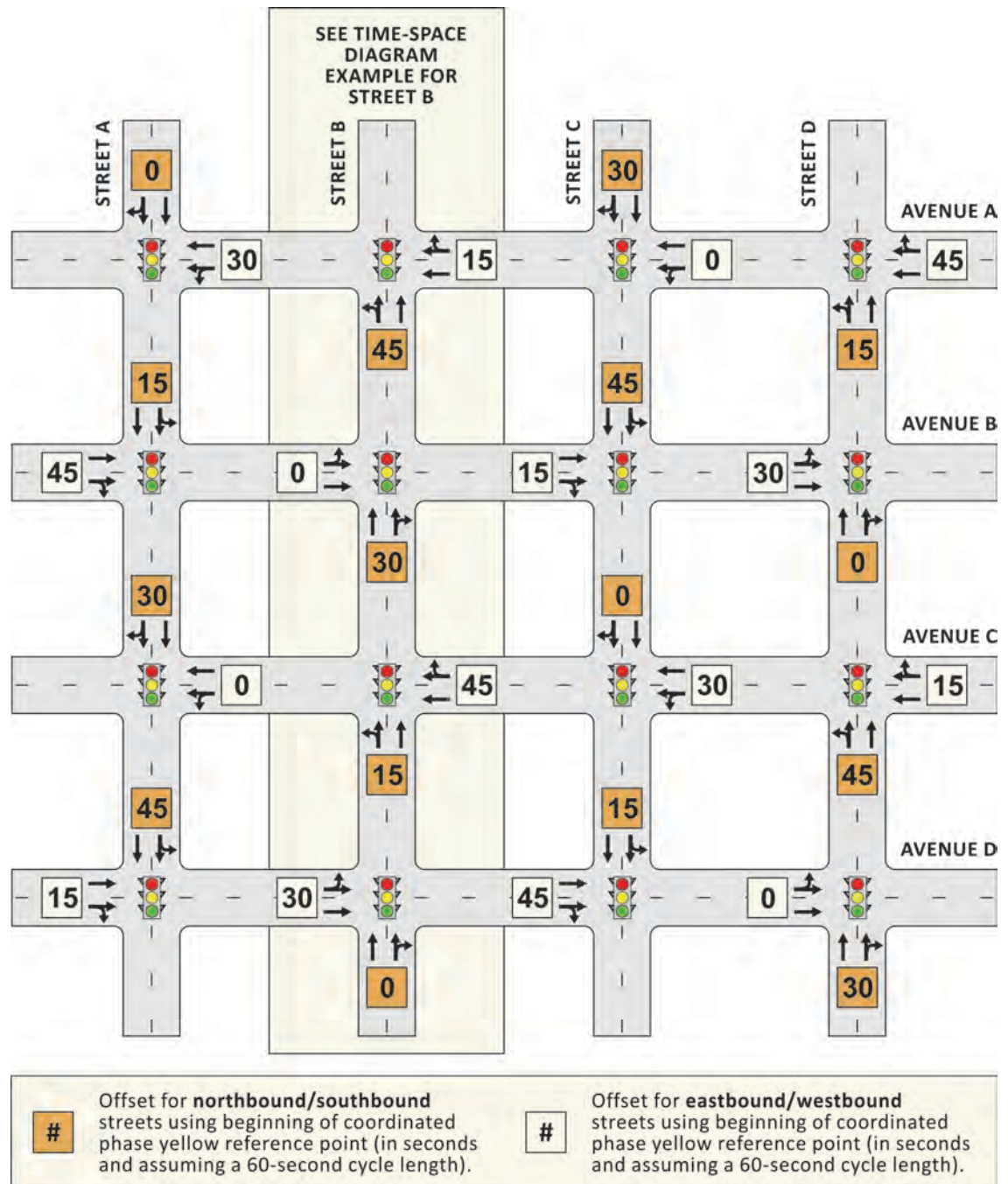


Exhibit 7-21 Triple
Alternate Coordinated
System

Manual methods for developing coordinated signal timing can also be applied to downtown grid networks. This method has been deployed in downtown Portland,

Oregon, by separating intersections into a quarter-cycle offset pattern (illustrated in Exhibit 7-22 and Exhibit 7-23). The block spacing in downtown Portland is fairly uniform and relatively short (280 feet), and the grid is a one-way network (resulting in each intersection operating with two phases). Each subsequent intersection is offset by a quarter of the cycle length, which was selected to progress traffic on both streets. The result is a 13 mph progression speed and a 60-second cycle length. Note that if a cycle length is short, pedestrian coordination can also be achieved in the opposite direction of the one-way vehicular movements.

Exhibit 7-22 Example of Quarter-Cycle Offset Coordinated System



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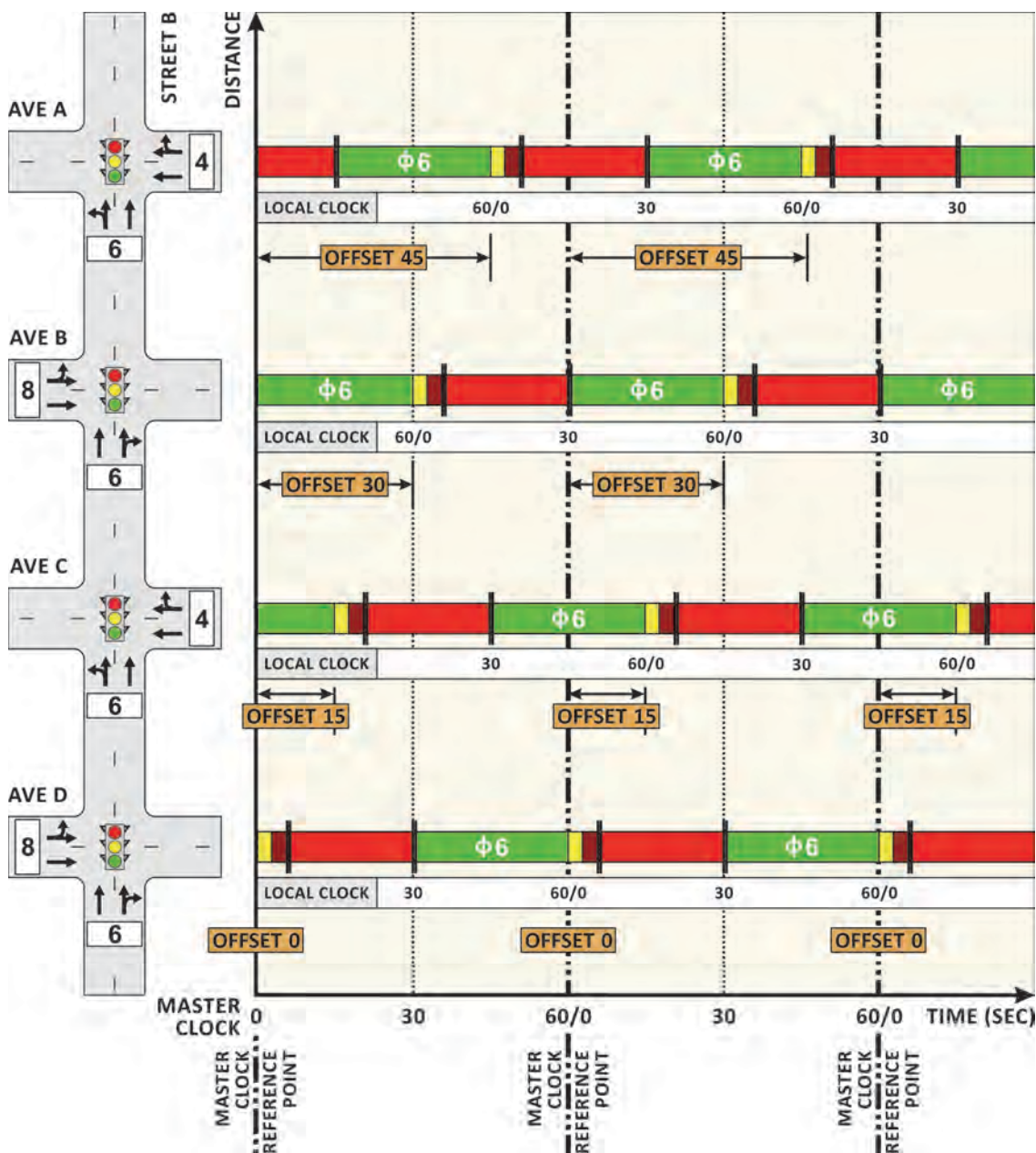


Exhibit 7-23 Example of Time-Space Diagram for Quarter-Cycle Offset Coordinated System

7.4.2.2 Critical Intersection Methods for Cycle Length Selection

As mentioned previously, the cycle length for a coordinated group of intersections can be based on the cycle length required at the critical intersection. Using this methodology, a cycle length is established that will sufficiently maintain under-saturated conditions at the critical intersection. While there are several critical intersection methods, the traditional method uses Webster's model (4) to determine optimal cycle length. The formula is as follows:

The critical intersection is typically the intersection with the highest demand.

Equation 7-1

$$C = \frac{1.5L + 5}{1.0 - Y}$$

where

C = optimum cycle length (seconds),

Y = critical lane volume divided by the saturation flow, summed over the phases, and

L = lost time per cycle (seconds).

The critical intersection approach considers signalized intersections in isolation, and for that reason, may not always yield the optimal cycle length for system progression (if that is the desired outcome). This method does not consider intersection constraints beyond the lost time and saturation flow rate and has a *de facto* outcome focused on vehicular flow. It does not specifically consider the needs of other users, such as pedestrians, bicycles, and transit, and is only valid for under-saturated conditions. (Oversaturated conditions require special consideration; more information can be found in Chapter 12.)

7.4.2.3 Network Approach to Cycle Length Selection

The network approach to cycle length selection considers the performance of multiple intersections when determining an optimal cycle length. Most applications use signal timing optimization models that are based on vehicular flow. The FHWA *Signal Timing Process Final Report* and *Traffic Analysis Toolbox* describe several available optimization software packages (5, 6). However, optimization models change with new versions of software, and, therefore, documentation is often best obtained from individual software producers.

Optimization models consider the network being analyzed and determine an optimal solution based on a given set of inputs (including the range of preferred cycle lengths). These models generally use individual intersection characteristics, volume-to-capacity ratios, link speeds, and the distance between intersections to estimate the performance of individual cycle lengths and resulting plans. The timing policies, optimization policies, and criteria used to select a signal timing plan must be considered prior to and as a part of the cycle length optimization process. A significant amount of effort may be required to take an initially screened plan from an optimization model to a point that can be field implemented.

7.4.3 Splits Guidance

Split distribution is the process of determining how much of the cycle can be provided to each of the phases. Determining adequate split times can be challenging. If a split time is too long, other phases may experience increased delay (if detection and associated passage time do not allow for termination at the end of maximum flow). If a split time is too short, the demand may not be served.

There are various ways to determine the necessary split time for a phase. The general intent is to allocate enough time to each phase to avoid oversaturated conditions for consecutive cycles, but over the course of an analysis period (15 minutes or 1 hour), splits should be distributed in a manner that is consistent with the operational objectives. For example, if the operational objective is minimizing stops on a

In many controllers, maximum green values may be ignored during coordination using the Inhibit Max feature. This allows a phase to use a split time that is longer than its normal maximum green value.

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corridor, the corridor phases should likely receive longer split times relative to the minor street phases.

One method for split distribution is allocating splits that satisfy a design level of capacity for all of the minor movements, with the remaining residual time allocated to the coordinated movements. Another common practice allocates the green time such that the volume-to-capacity ratios for critical movements are equal. With the many interactions between signal timing parameters, the allocation of green time should also depend on pedestrians, transit phases, and bandwidth considerations.

7.4.4 Force-Offs Guidance

Controllers often automatically calculate force-offs based on the programmed splits. However, some controllers may allow the practitioner to program the force-off values, or the type of force-off that is applied. The phase directly after the coordinated phase will not have an opportunity to receive time from a preceding phase, regardless of the method of force-off (unless the coordinated phase is actuated). Thus, the force-off (or split) for the first phase following the coordinated phase should be selected carefully.

Depending on the operational objectives, consideration should also be given to the type of force-off that is applied. Floating force-offs favor the coordinated phase(s), as they do not allow uncoordinated phases to inherit time. Fixed force-offs, on the other hand, can be beneficial if there are fluctuations in traffic demand, and an uncoordinated phase needs more green time during a cycle. This type of force-off can also help to prevent early return to green on the coordinated phases (7), reducing perceived delay along the corridor (described in detail in Section 7.6.2).

One of the potential outcomes of fixed force-offs is that a phase later in the sequence (before the coordinated phase) may receive more than its split time (provided the maximum green timer is either not active or is not reached). Inhibit Max (a definable controller parameter) may be invoked to prevent the controller from using maximum green values during coordinated operations. If allowed to function during coordination, maximum green values could result in phases not reaching their force-off points.

Inhibit Max must only operate during coordinated time periods.

7.4.5 Permissives Guidance

Many modern controllers automatically maximize permissives, but they do not all operate in the same manner, as discussed earlier. Larger permissive periods are desirable during times of day with low traffic volumes because of the increased opportunities for uncoordinated phases to be served. With higher traffic volumes, permissives are not a significant issue because a call usually exists on all phases when the controller reaches the yield point. Maximizing permissives is generally recommended.

7.4.6 Yield Point Guidance

Yield points are ultimately determined based on when permissive periods begin. Modern controllers calculate yield points based on the offset reference and walk mode. No other consideration is necessary.

7.4.7 Pattern Sync Reference Guidance

Pattern sync reference should generally be programmed at a time when there are lower traffic volumes in order to limit disruptions to traffic flow. It is important that each intersection reference a consistent master clock so that local controllers use identical reference points. Each controller should be configured to keep track of subtle time-related issues that will keep it in sync with the master clock, such as if the area follows daylight savings time and/or when daylight savings time begins and ends. If the local clock time is not synchronized with the master clock, coordination will not function correctly.

7.4.8 Offset Reference Point Guidance

Offset reference points will not move based on when phases actually time; they reference a consistent point in time that is based on phase splits and sequence. Reference points are generally programmed through controller or system firmware, and it is essential for the practitioner to apply a consistent offset reference point for a group of coordinated signals. However, it is important to realize that all offset reference points are equivalent operationally (i.e., if you know one reference, you can calculate any other reference for the same pattern).

The beginning of first coordinated phase green has been a common offset reference point and is the official value defined in NTCIP 1202 (2). However, it is not readily observable in the field because uncoordinated phases can terminate early through gap outs. This causes the coordinated phases to begin earlier than the programmed “beginning of first coordinated phase green” offset reference point.

The beginning of first coordinated phase yellow is generally observable or apparent in the field (i.e., the actual point does not move as a result of uncoordinated phases timing less than their allocated splits). One exception to an observable beginning of yellow is if the coordinated phases are actuated. In that case, only the beginning of FDW is observable (although not a typical offset reference).

7.4.9 Offsets Guidance

Offsets should be chosen based on the actual or desired travel speed between intersections, distance between signalized intersections, and traffic volumes. In an ideal coordinated system, offsets would allow platoons (leaving an upstream intersection at the start of green) to arrive at a downstream intersection near the start of green or after the queue from minor streets or driveways is discharged (i.e., green starts early enough to clear queued vehicles before platoon arrives).

Offsets are often adjusted in the field, but field observations provide the practitioner with a limited ability to review conditions. With a 100-second cycle, there are only 36 cycles during an hour, and one cycle may not be indicative of the next cycle’s performance. Instead of attempting to base offsets solely off of field observations *or* software output, the practitioner should use field review *and* time-space diagrams in combination to optimize the system. Some modern controllers also monitor arrivals on green as an aid to fine-tuning offsets.

The examples throughout this manual use the beginning of first coordinated phase yellow as the offset reference point.

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7.5 OTHER CONSIDERATIONS FOR COORDINATION

Beyond basic coordination parameters, there are other considerations for coordinated operations. A practitioner should understand how walk modes, actuating the coordinated phase(s), and transition logic will affect operational objectives.

7.5.1 Pedestrian Timing and Walk Modes

Pedestrian operations can have a direct impact on the coordination along a corridor. Pedestrian timing is required for all phases that serve pedestrians. However, when pedestrian service is actuated and demand is relatively low, it may be desirable to allocate a split time that is shorter than the time required to serve a pedestrian (if the controller firmware supports the capability). Traffic operations may be more efficient without accommodating pedestrians within the coordinated cycle length (if the pedestrian time is longer than what is needed for vehicular progression and traffic demand). It may be more effective for a controller to be shifted out of coordination and have to transition back for the occasional pedestrian than to serve pedestrian timing every cycle.

7.5.1.1 Pedestrian Timing for Uncoordinated Phases

The time necessary to walk across the street (pedestrian timing requirements) may be longer than the needs of other traffic. The effect of pedestrian timing on coordination is, therefore, most apparent as it affects the timing on the minor street. Exhibit 7-24 illustrates the basic principle of pedestrian timing when the vehicle split is sufficient to accommodate the required pedestrian time, allowing the signal to stay in coordination (and not exceed the force-off point).

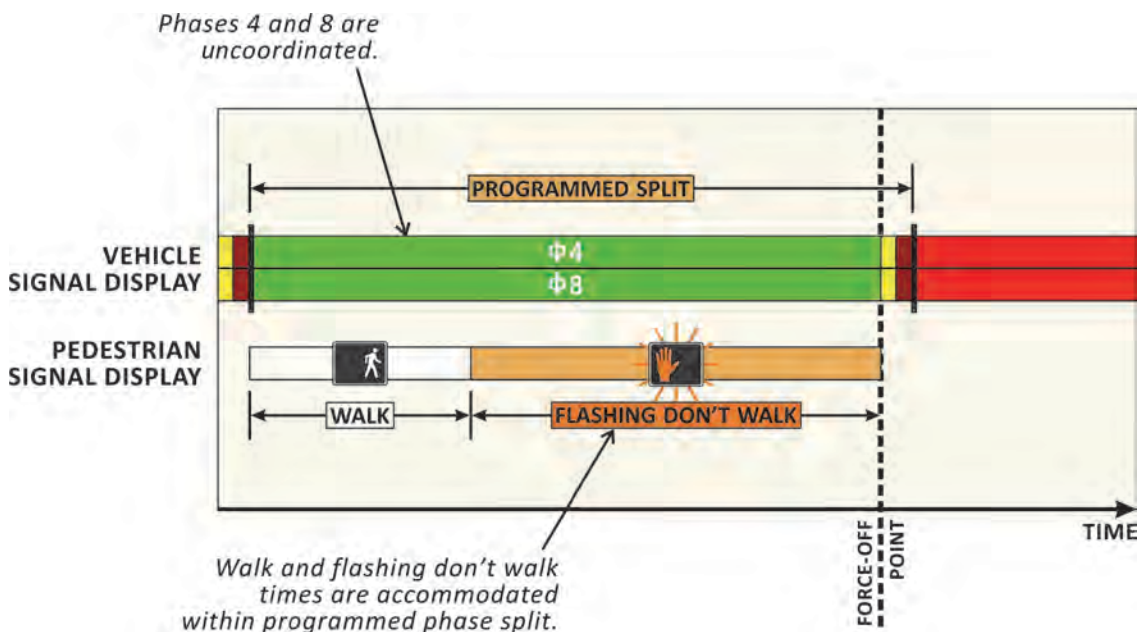


Exhibit 7-24

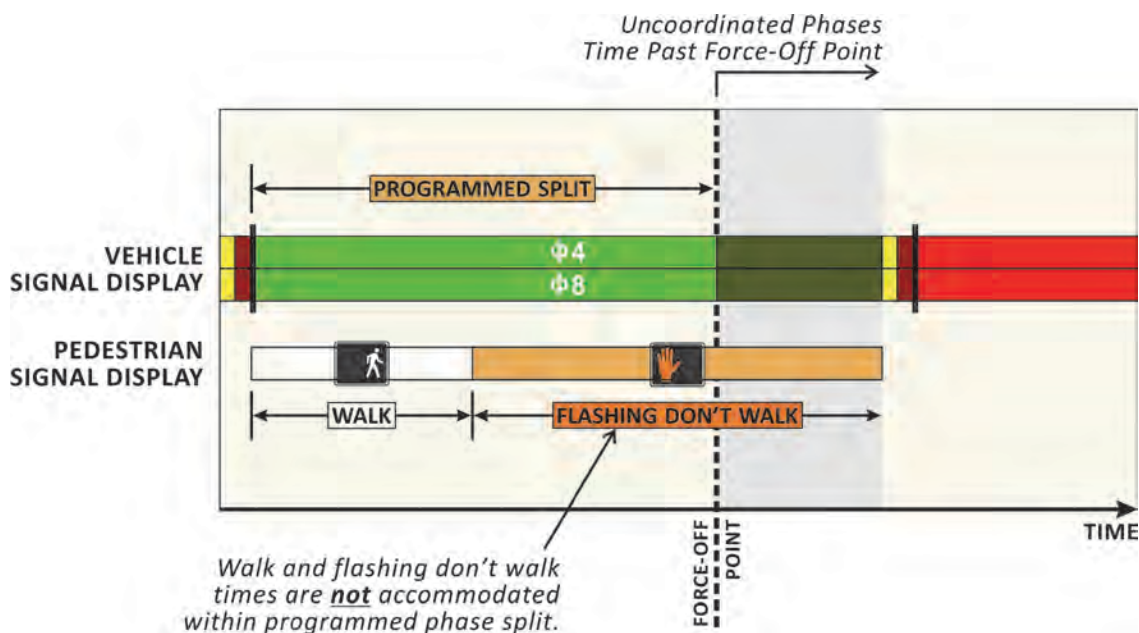
Uncoordinated Phase Operation with Pedestrian Timing Completed Before the Force-Off

When the split for the subject phase is not sufficient to cover the pedestrian timing, the controller times the phase beyond its force-off point (as illustrated in Exhibit 7-25). The response of the controller depends on two factors: (1) demand for subsequent

uncoordinated phases and (2) non-actuated versus actuated operations on the coordinated phases.

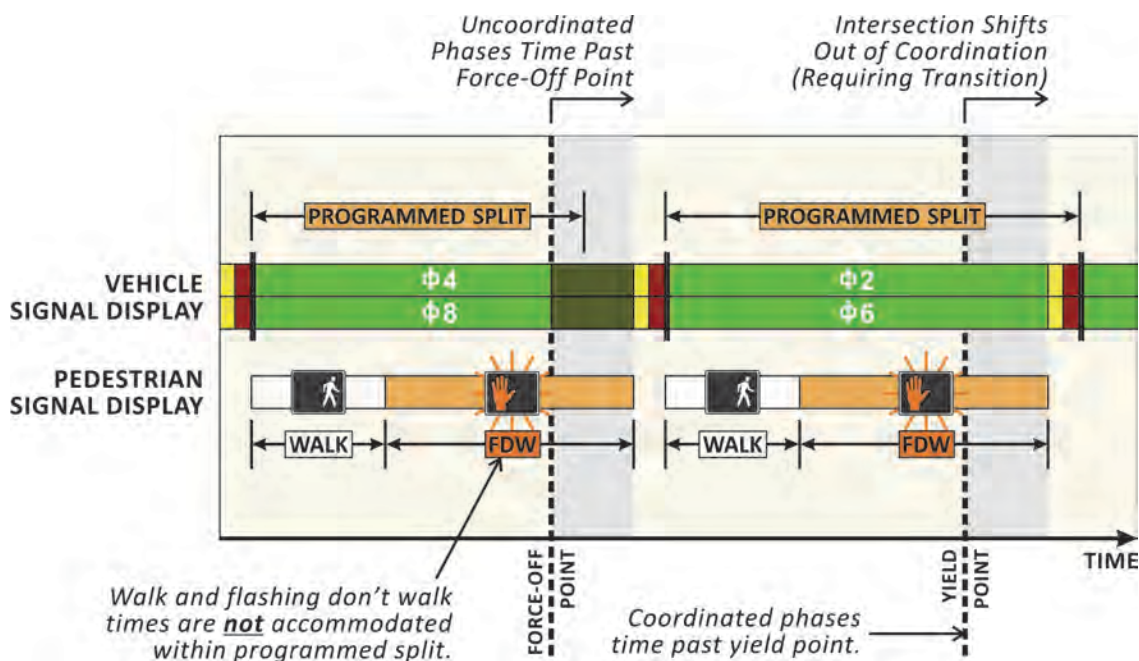
Exhibit 7-25

Uncoordinated Phase Operation with Pedestrian Timing Exceeding Phase Split



When the coordinated phases are non-actuated (which is typical practice), the coordinated phases must begin timing sufficiently in advance of the controller's yield point to enable full vehicle timing (minimum green) and pedestrian timing (walk plus FDW). Should the amount of time be insufficient to cover these timing requirements, the controller will time the coordinated phase past the yield point and fall out of coordination (as shown in Exhibit 7-26).

Exhibit 7-26 Loss of Coordination Due to Pedestrian Call



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It is at the yield point that the controller logic typically determines the method by which the controller will transition back into coordination. If the late return to the coordinated phase is minor (relative to cycle length and requirements for other phases), it could provide minimal disruption and significant benefits to overall traffic, provided the transition mode allows the controller to take time from following phases (see Section 7.5.3).

7.5.1.2 Pedestrian Timing for Coordinated Phases

The amount of time needed to serve vehicular volumes or provide bandwidth along the major street usually results in coordinated phase splits that are sufficient to accommodate pedestrian timing. While progression considerations generally provide adequate time to accommodate pedestrians, some walk modes are more pedestrian-friendly than others, as discussed below.

7.5.1.3 Walk Modes

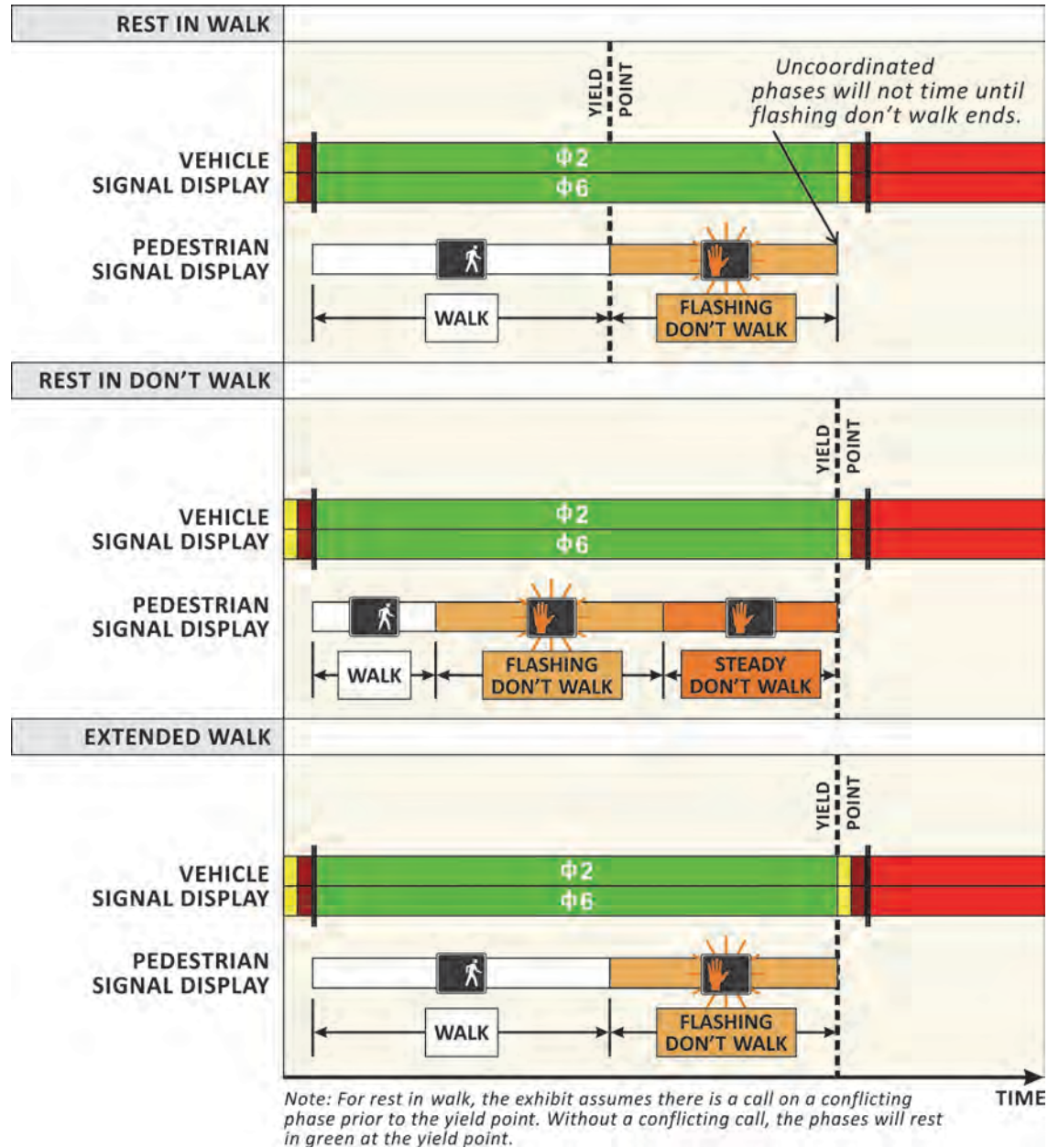
A practitioner can specify walk modes that influence how pedestrians are served during the coordinated phases: (1) rest in walk, (2) rest in don't walk, and (3) extended walk (Exhibit 7-27 illustrates these three walk modes):

- **Rest in Walk** dwells in the pedestrian walk interval while the coordinated phase is green, regardless of pedestrian calls. This mode is often used when there are high pedestrian volumes, such as in downtown environments or locations near schools, and it does not **require** any pedestrian detection (although pedestrian detection may be desirable to allow for late-night free operation). However, this walk mode causes the FDW interval to extend past the yield point, delaying minor street movements until the FDW interval has ended. The delay to minor streets is only noticeable under low-volume conditions.
- **Rest in Don't Walk** dwells in the steady don't walk interval after the programmed walk and FDW intervals have been served. This mode is often used when pedestrian volumes are low. It does require pedestrian detection.
- **Extended Walk** dwells in the pedestrian walk interval (similar to rest in walk) starting at the beginning of the coordinated phase green. It maximizes the walk time every cycle, but also times the FDW interval prior to the yield point. This mode is a compromise between the first two walk modes, and it does not require pedestrian detection.

The appropriate walk mode may depend on the time of day. During the middle of the night, rest in don't walk might be the most appropriate because of the low volume of pedestrians. However, a high volume of pedestrians during the day may make rest in walk (or extended walk) the most appropriate for that time period.

Depending on the walk mode and pedestrian actuation settings, the walk interval may be able to time more than once during a cycle. Pedestrian re-service is a feature that can be used when pedestrian service is actuated on the coordinated phase. It allows the walk interval to time again if (a) a call is placed and (b) there is enough time before the latest point at which the FDW interval must begin.

Exhibit 7-27
Pedestrian Walk
Modes



7.5.2 Actuating the Coordinated Phase

A portion of the split time at the end of the coordinated phases can be actuated, allowing for an earlier termination of the coordinated phases in the absence of detected traffic demand (as illustrated in Exhibit 7-28, assuming rest in don't walk). This approach effectively moves the phase termination point earlier in the cycle and allows the coordinated phases to gap out and give time to the minor streets and left turns. In addition, it increases the likelihood of decision zone protection (through the use of detection to end the coordinated phases instead of a max out or force-off).

An actuation value that is just right will (1) allow for occasional gap outs to serve minor street demand more quickly (reducing intersection delay) and (2) reduce max outs or force-offs for the coordinated phases (enhancing decision zone protection). An

A portion of the coordinated phase can be actuated, which increases the likelihood of decision zone protection and offers more flexibility to serve minor street and left-turn movements.

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“actuated time” value that is too large can result in more early returns to green and/or the coordinated movements gapping out just prior to the arrival of the platoon, resulting in poor signal progression, increased stops, and delay along the corridor. A value that is too small will have an inconsequential result. Actuating the coordinated phase requires engineering judgment and observation of signal operations during multiple times of day or days of the week to set reasonable actuation values (typically per ring).

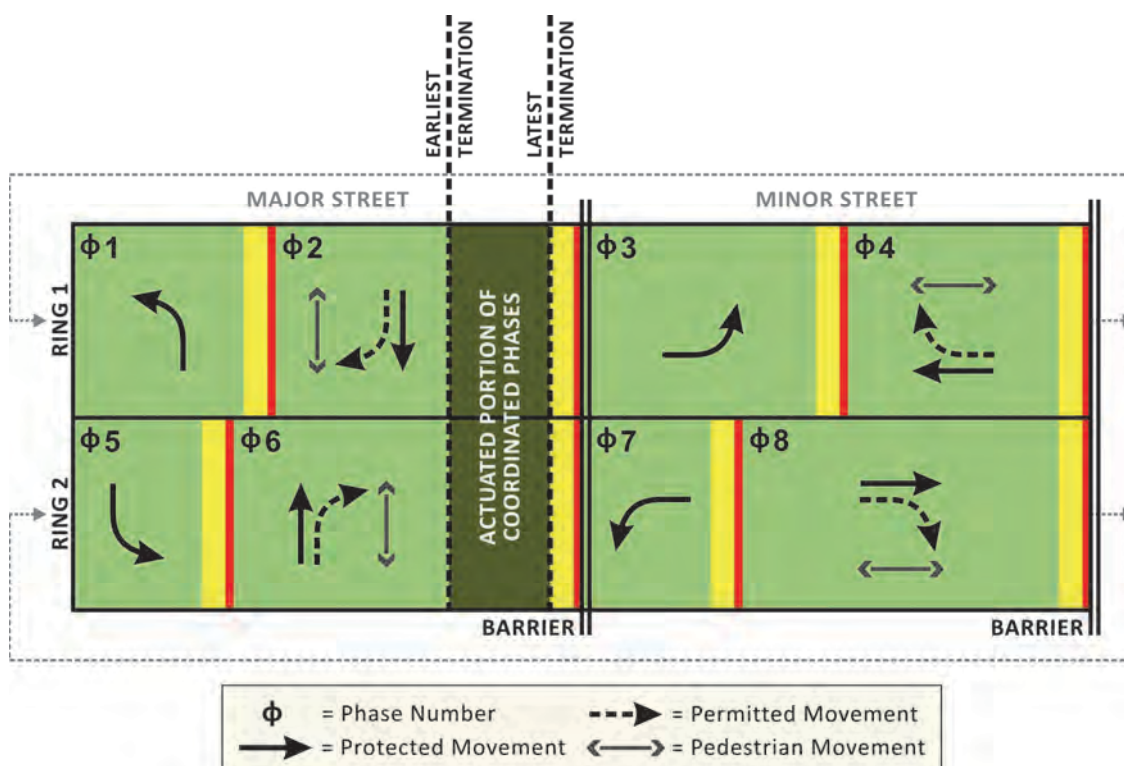


Exhibit 7-28 Actuating a Portion of the Coordinated Phases

7.5.3 Transition Logic

Transitioning is the process of either entering into a coordinated timing plan from “free” operations or changing between two plans. Transitioning may also be necessary after an event such as preemption or loss of coordination (possibly due to a pedestrian time that exceeds the allocated split, as discussed in Section 7.5.1). In general, traffic signals do not operate within the same pattern parameters and cycle lengths at all times. The pattern may change during the day for a number of reasons:

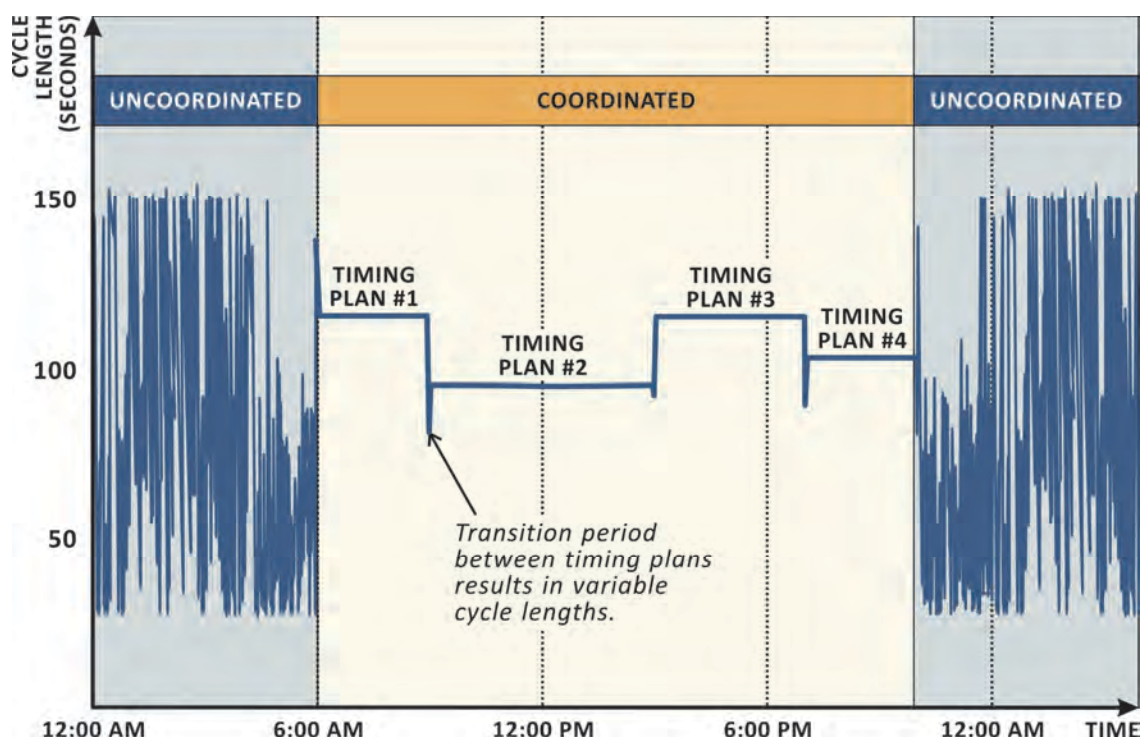
- Time-of-day scheduled changes;
- Manual operator selection;
- Traffic-responsive pattern selection;
- Emergency vehicle, railroad, or other preemption;
- Adaptive control system pattern selection;
- Corrections to the controller clock;
- Pedestrian time exceeds split time; or
- Power loss and restoration.

Studies have shown that excessive changes to timing plans, in an attempt to match traffic patterns closely and improve performance, can be a detriment because the system never achieves coordination long enough for the new plan benefits to outweigh the costs of transition. Because of this, it is generally recommended to remain in a coordinated pattern for at least 30 minutes. It is also best to avoid changing patterns during congested conditions when the signals need to operate at maximum efficiency. A peak-period pattern is best implemented early to ensure all offset transitioning is completed before the onset of peak traffic flows.

When the controller reaches a point when it is necessary to change the coordination pattern (the cycle, splits, offsets, and/or sequence), the controller may also have to shift the local offset reference point. This requires the use of an algorithm that may either shorten (i.e., subtract time from) or lengthen (i.e., add time to) the cycle. The transition algorithm typically operates over one to five cycles, depending on the transition mode selected and how much the offset reference point needs to shift. Consequently, the split durations (and cycle lengths) during the period of transition will be different from those programmed in either the previous pattern or new pattern.

For example, in the cycle length plot shown in Exhibit 7-29, the system runs with a fixed background cycle from 6:00 a.m. to 10:00 p.m., with cycle changes at 9:00 a.m., 3:00 p.m., and 7:00 p.m. Before 6:00 a.m. and after 10:00 p.m. the intersection is running under free (uncoordinated) operations, so the cycle length varies depending on the traffic detected at the intersection. During each of the plan changes, the controller goes into transition, resulting in variable cycle lengths for a few cycles to adjust to the new pattern. These variable cycle lengths are portrayed by the dips and rises in the overall steady cycle length line.

Exhibit 7-29 Example of Daily Cycle Length Fluctuations



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While central- or master-based systems typically communicate the selection of a new timing plan to all signals in a group at the same time, the actual transition logic is typically executed independently at each signal, without explicit regard for the state of adjacent signals. Most controllers allow three or four transition modes, which govern the precise details of how the signal resynchronizes to the new offset reference point. The transition modes differ significantly from one controller manufacturer to the next. Some vendors may refer to transition as offset seeking, offset correction, or coordination correction (8). No matter which mode is selected, traffic control can be significantly less efficient during the transition between timing plans than it is during coordination.

The three most common techniques for achieving an offset reference point transition are

- Lengthening the cycle (adding time to each split),
- Shortening the cycle (subtracting time from each split), and
- Shortway (adding or subtracting time depending on which is faster).

To avoid a cycle length that is excessively long or green times that are too short during the transition period, it is common for controllers to limit the maximum amount of adjustment that can be made in one cycle. If such a limit is imposed (which is typical), the signal may not be able to complete a given transition within one cycle even if the adjustment is small. Signal controllers also commonly compute their adjustments so that transitioning is completed in a set number of cycles for the worst case scenario, typically a maximum of three to five cycles.

7.5.3.1 Lengthening Transition Modes

The most common lengthening transition modes provided by signal controllers in the United States include dwell, max dwell, and add only (which is generally preferable to dwell modes) (9). Lengthening transition modes offer less risk than transition modes that shorten the cycle length, unless the amount of shortening is known to be small. There is less likelihood of queues building up because of short green times and pedestrians not being served. However, if the offset reference point is being shifted 1 second backwards, using a lengthening transition mode requires shifting the entire cycle forward 1 second less than the cycle length. This results in longer cycles during the transition period, which could potentially cause unexpected storage problems in left-turn lanes or between closely spaced intersections. The modes are described as follows:

- **Dwell:** At the next display of green for the coordinated phases, the controller begins to transition by holding (or dwelling) in this state until the new offset reference point is achieved, at which time the signal is considered in sync and begins the new timing plan. This transition mode puts all the transition time into the coordinated phases, which may cause problems on other uncoordinated phases. If the offset reference point needs to move 1 second earlier, the coordinated phases will dwell for 1 second less than the cycle length.
- **Max Dwell:** This modified version of dwell also adjusts the start of the cycle by extending the green time of the coordinated phases. However, only a limited amount of extra green time may be added each cycle.

- **Add:** This mode synchronizes by shifting the start of the cycle progressively later, by timing slightly longer-than-programmed cycle lengths. The add mode increases the green time on all phases in the sequence, whereas the dwell modes add time only to the coordinated phases. If a signal is subject to preemption, selecting the add-only transition mode allows all phases to receive additional time during the transition period.

7.5.3.2 Shortening Transition Mode

The shortening transition mode shortens the cycle length, subtracting time from phases to the extent allowed by their minimum green settings and any pedestrian activity during the phase. Shortening can be effective if the offset correction is very small. Depending on the minimum green times and pedestrian times, there may only be small adjustments in cycle length that can be made by shortening. It may take many cycles to complete an offset reference point transition, so it is not practical to require a controller to use the shortening technique exclusively. To avoid this problem and allow use of the shortening transition mode when it works well, most controllers offer some version of the shortway method (discussed in the next section).

- **Subtract:** This mode shifts the start of the cycle progressively earlier, subtracting time from one or more phases in the sequence (subject to their minimum green time and pedestrian time requirements). This method is very effective for small corrections, such as pedestrian calls that go past the force-off by a small amount.

7.5.3.3 Shortway Transition Mode

If the practitioner selects the shortway transition mode, the controller will assess both lengthening and shortening techniques and automatically choose the one that will complete the offset reference point transition (i.e., get the signal "in step" or "in sync") most quickly. Field experience and laboratory experiments have shown that shortway typically provides the least disruptive effects on traffic compared to other methods, unless the signal is subject to preemption or the intersection is at near-saturated conditions. "Traffic Signal Transition in Coordinated Meshed Networks" reports that shortway provides quick and smooth transitions while avoiding severe peaks in delay. While this method was sometimes outperformed by others when a single approach was considered, it was always among the best methods in overall performance (10).

- **Shortway:** This mode (also sometimes called bestway, fastway, or smooth) finds the "shortest path" to transition the intersection by using either lengthening or shortening transition logic. The specific details of how this mode determines the "shortest path" can vary significantly from one controller vendor to the next.

7.6 COMPLEXITIES

This section discusses some of the various complexities of signal coordination. There are many variables that must be considered to achieve an acceptable coordination plan.

Shortway is typically the least disruptive method that can be used to achieve an offset reference point transition.

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7.6.1 Phase Sequence

The sequence of phases, particularly left-turn phases, can significantly affect corridor operations. The most common phase sequencing decision—whether to lead or lag left turns—can have a particularly strong impact on bandwidth (in both directions) along a corridor. Other phase sequence decisions (such as the sequence of left turns on the minor street or the sequence of split phasing on the minor street) often have less impact on bandwidth and delay but should also be considered.

7.6.1.1 Major Street Left-Turn Phase Sequence

Lagging one of the major street left-turn phases (lead-lag) can facilitate better progression for both directions because it allows platoons to arrive at different times during the cycle (as demonstrated in Section 7.2.6). Depending on how the lagging left-turn phase is configured, it will generally receive the same amount of green time each cycle regardless of demand (unless the coordinated phase in the other ring has an actuated interval, discussed in Section 7.5.2). This fixed interval occurs because the coordinated phase in the other ring needs to end at the same time as the lagging left in order for both rings to terminate and cross the barrier together. Actuating the coordinated phase allows both the coordinated phase and lagging left to end early if demand is not present.

Modern controllers allow left-turn phase sequences to be varied by time of day. This has traditionally been done only for protected left-turn operations, but the use of FYA indications allows this to be extended to protected-permitted operations (see Chapter 4 for more information on FYAs). The practitioner should always consider user expectations and the operational objectives when choosing different phase sequences for different times of day.

7.6.1.2 Minor Street Left-Turn Phase Sequence

It may be advantageous in some circumstances to adjust the protected left-turn phase sequence for the minor street. In doing this, it may be possible to reduce the delay and queuing for minor street left turns as they enter the major street and arrive at downstream intersections. Such adjustments (which require an optimization tool that shows minor street platoons) may affect system-wide delay, stops, and bandwidth on the corridor.

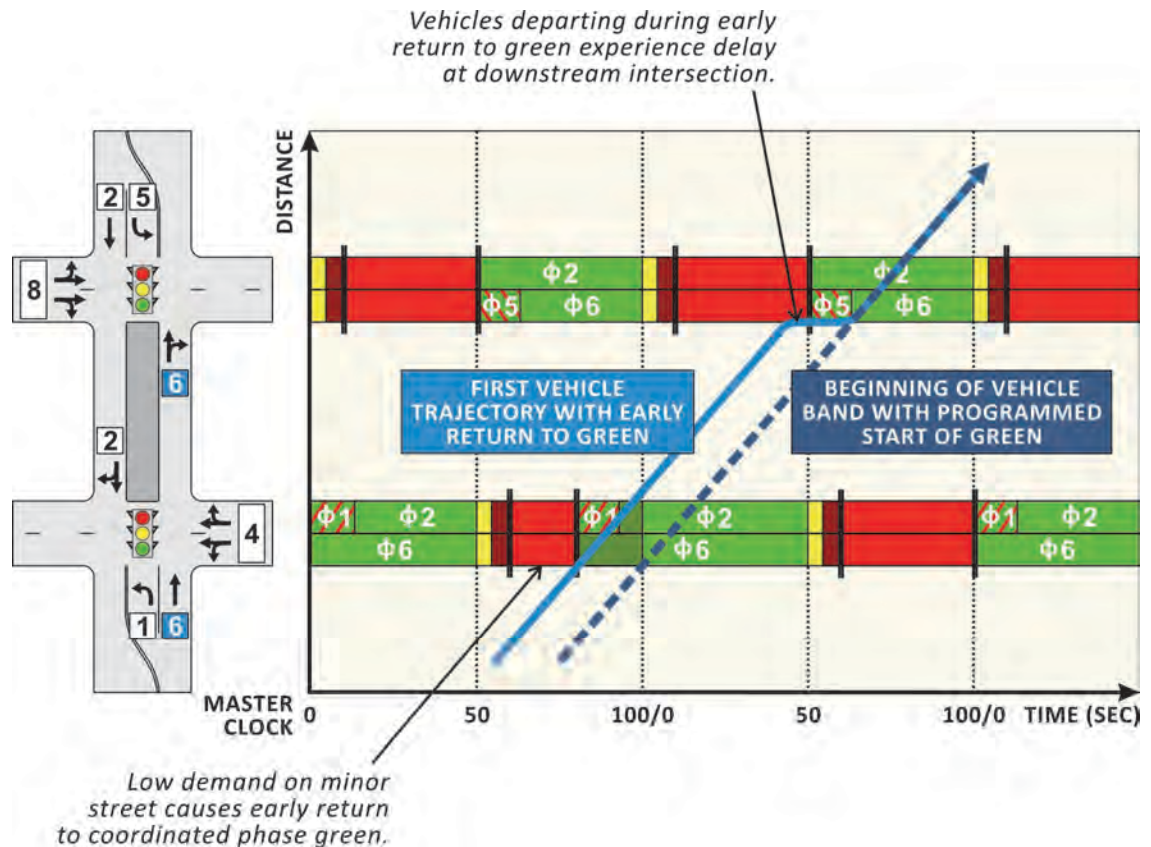
7.6.2 Early Return to Green

One of the consequences of actuating the uncoordinated phases is the potential for the coordinated phases to begin earlier than expected. This “early return to green” occurs when the sum total of the time required by the uncoordinated phases is less than the sum total of the vehicle splits programmed for the phases. While this may reduce delay at an intersection, it may increase stops at downstream intersections, which is perceived poorly by users. However, if early release does not discharge vehicles or releases queued vehicles ahead of an arriving platoon, it may have positive consequences. It is, therefore, necessary to review several intersections from a system perspective to determine the effects of early return to green.

Exhibit 7-30 illustrates early return to green within a time-space diagram. The figure shows that if the coordinated phases begin early, vehicles may be forced to stop

at one or more downstream intersections until they fall within the “band” for that direction of travel. This can result in multiple stops for vehicles and a perception of poor signal timing. Early return to green can be difficult to manage along a corridor, and it can rarely be completely prevented without eliminating most of the benefits of actuation. One technique that can be used is to delay the start of the coordinated phases (i.e., shift the intersection offset to account for the expected early green) if there is a high probability of early return to green. For example, early return to green may be probable if there is a large minor street split (possibly programmed to accommodate an occasional pedestrian call) that is often not fully utilized.

Exhibit 7-30 Example Time-Space Diagram Showing Early Return to Green



7.6.3 Heavy Minor Street Volumes

Heavy minor street volumes can affect the ability to progress through movements along a corridor. These minor street volumes can come from signalized intersections within the coordinated signal system, unsignalized intersections, or driveways between coordinated signals. Interchanges are also a common source of heavy minor street volumes.

In many cases, this additional demand proceeds along the remainder of the corridor and becomes part of the major street through demand at downstream intersections. However, this demand often enters the system outside the band established for through movements traveling end-to-end along the corridor. It may be desirable to adjust downstream intersection timing to allow these heavy minor street movements to proceed with a minimum number of stops, unless it is a minor street with a likely early return to green.

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7.6.4 Turn-Bay Interactions

Turn-bay (or turn-pocket) interactions can impact the effective capacity of an intersection. This is experienced when either demand for the turning movement exceeds the available storage space or when vehicle queues block the entrance of a turn bay. Turn-bay overflows can adversely impact progression by disrupting through traffic from proceeding to downstream intersections or altering the arrival profile. This reduces the ability for downstream intersections to efficiently provide green time for the platoon. Beyond being aware of the effects of turn-bay interactions on coordination, practitioners can find techniques to mitigate this type of oversaturation in Chapter 12.

7.6.5 Critical Intersection Control

A challenging aspect of timing an arterial street or a network of streets is the need to provide enough capacity for major intersections without creating excessive delay for minor intersections. Ideally, all of the intersections to be coordinated operate optimally with similar cycle lengths. However, most arterial streets do not have this optimal arrangement due to a mixture of minor intersection signals (e.g., no left-turn phases) with more complex signals (e.g., eight phases), wide ranges in cross-street volumes (e.g., major arterials versus collectors), and variations in left-turn volumes. Several techniques can be used in situations where there is a significant disparity in the ideal cycle length:

- Each intersection is timed using the critical intersection cycle length. This ensures the ability to coordinate all of the intersections in the system. However, use of this technique may result in excessive delay at minor intersections.
- Each intersection is timed to either the critical intersection cycle length or to half that value. This technique is commonly referred to as “double cycling” (i.e., a minor intersection cycles twice as frequently as a major intersection) or “half cycling” (i.e., a minor intersection has half the cycle length of the major intersection). It is also possible in some controllers to have two unequal (asymmetrical) cycles by not providing all phases or constraining splits. These methods can often produce substantially lower delays at the minor intersections where double cycling is employed. However, it may become more difficult to achieve progression in both directions along the major arterial, which could result in more arterial stops than desired.
- The major intersections are operated freely, and the minor intersections are coordinated using a shorter cycle length. Because the major intersections are operating freely, a traditional coordination band is impossible. Therefore, major street vehicles are likely to stop at both the major intersection and at a downstream intersection due to randomness in arrival at and departure from the major intersection. This technique can often result in lower overall system delay at the expense of additional stops along the major street. However, with splits neither constraining green allocation nor driving up cycle length due to pedestrian times, the critical intersection may have fewer phase failures.

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